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Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV

By
Abdul Ahad

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Abstract

Over ninety AI models have been built to predict sepsis in real time, yet they fail in practice. Wang et al. (2025) showed that while AUROC barely changes between internal and external testing (0.811 to 0.783), the clinical Utility Score collapses from +0.381 to -0.164 , meaning these models cause more harm than benefit when deployed. This thesis investigates why: is the problem the model, the sepsis label, or the researcher’s methodological choices?

The study uses MIMIC-IV v3.1 (65,078 adult ICU stays) with external validation on eICU-CRD v2.0 (63 sepsis cases across 7.9 million person-hours). Three Sepsis-3 label variants (Narrow, Seymour-Standard, Liberal) are tested with a dynamic discrete-time competing-risks hazard model, gradient-boosted trees (GBT), a static Cox model, and rule-based scores (NEWS2, qSOFA, SIRS). Because all three variants share one culture-plus-antibiotic anchor, the Sepsis-3 conjunction is additionally split into its two limbs and each is labelled separately, including a deterioration label that uses no treatment timestamp at all.

The results show that the sepsis label definition changes AUROC by 0.153 (95% CI 0.136–0.177), a spread statistically indistinguishable from the one produced by changing the model (0.142, 95% CI 0.130–0.161): both analyst decisions are large and neither dominates. Ten of the twenty shared clinical covariates reverse direction across labels, making published risk factors unreliable without knowing which label was used. Random data splits inflate AUROC by 0.123 (95% CI 0.100–0.147) compared to time-based splits. Most importantly, the pre-treatment evaluation window contains **zero sepsis events** in 179,003 person-hours: no patient develops sepsis before doctors begin treatment. This means current models predict when clinicians act, not when patients deteriorate. The Utility Score is **negative for every model and every label**, with over 90 false alerts for each true one.

Splitting the definition shows how much of the label effect is definitional and how much is real. The onset *time* under every Sepsis-3 variant is fixed entirely by the treatment anchor, the organ-dysfunction criterion selecting which stays are labelled without moving any onset by an hour. Yet a treatment-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance, so the label effect

is not an artefact of re-reading one definition. Sepsis-3 also proves *harder* to predict than either limb it is built from — 0.607 AUROC against 0.756 for the treatment anchor alone and 0.842 for deterioration alone — which places the difficulty in the conjunction rather than in the physiology or in clinician behaviour. Critically, the same deterioration label records **1,362 onsets in the pre-treatment window where Sepsis-3 records none**, at 0.840 AUROC: patients do deteriorate before clinicians act, and the empty window is a limitation of the definition rather than a fact about the patients.

All estimates carry stay-level cluster-bootstrap intervals; the six pre-registered hypotheses are corrected by Holm–Bonferroni and all subgroup analyses by Benjamini–Hochberg. Under correction, subgroup discrimination differences by race are too imprecisely estimated to support any claim, while differential *label* sensitivity does survive: patients whose race is undocumented, and Spanish-speaking patients, change sepsis status across label definitions significantly more often than the reference groups.

The collapse in clinical utility is not a model problem; it is a label problem. Better models cannot fix a label that depends on the actions it is supposed to predict ahead of.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; competing risks; label definition; variance decomposition; clinical utility; temporal dataset shift; treatment-anchored labels; TRIPOD+AI.

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I am grateful to my supervisor for guidance throughout this research and to the Atlantic Technological University (ATU) for the opportunity to pursue this work.

This research uses the MIMIC-IV Clinical Database (v3.1, Johnson et al., 2023) and the eICU Collaborative Research Database (v2.0, Pollard et al., 2018), both made available through PhysioNet under credentialing agreements. I thank the patients whose de-identified records form these databases, the clinical staff who generated them, and the BIDMC/MIT/PhysioNet teams who maintain them.

The open-source R and Python communities, in particular the maintainers of `data.table`, `arrow`, `nnet`, `xgboost`, `sandwich`, and `pmsampsize`, are acknowledged for the packages that made this pipeline possible.

Chapter 1

Introduction

1.1 Context and Motivation

Sepsis is a life-threatening condition caused by the body’s extreme response to infection [1]. It kills about eleven million people every year, nearly one in five deaths worldwide [2]. In the ICU, every hour of delayed treatment raises the risk of death [3]. An early warning system that detects sepsis before doctors recognise it could save lives by starting treatment sooner.

This potential has driven major research investment. Wang et al. [4] reviewed ninety-one published real-time sepsis prediction models and found a median internal AUROC of 0.811, suggesting strong performance. But the same review found that the clinical Utility Score collapsed from +0.381 internally to −0.164 externally. A negative score means the model’s alerts cause more harm than benefit. AUROC did not detect this collapse because it only measures whether a model can rank patients correctly, not whether its alerts are timely or useful. This gap between apparent accuracy and real-world usefulness has also been observed in deployment: Wong et al. [5] evaluated the Epic Sepsis Model v2 across 227,091 encounters and found high AUROC alongside an alert burden that clinical staff described as unmanageable.

The key question is therefore not “which model is best?” but something more fundamental: *how much of a model’s reported performance comes from the model itself, and how much comes from how the study was designed: how sepsis was defined, how data was split, and which metric was reported?*

Cohen et al. [6] showed on MIMIC-III that varying only the Sepsis-3 definition changed the identified cohort from 867 to 2,178 patients, and that the label choice affected performance as much as the model choice. Kamran et al. [7] showed that published models partly learn to predict when clinicians start treatment rather than when patients deteriorate. Together, these findings suggest that much of the

reported performance may come from the researcher’s design choices, not from the AI model.

This thesis tests that hypothesis directly. Instead of fixing the label, the split, and the metric and asking which model wins, it treats all three as **sources of variance** and measures how much each one contributes to the spread in reported performance.

1.2 Limitations of the Conventional Approach

The standard approach to sepsis prediction follows a simple formula: choose one Sepsis-3 definition, use one train/test split, report AUROC, and compare models. Most of the ninety-one models reviewed by Wang et al. [4] follow this pattern.

This approach has four key problems:

1. **The sepsis label is not fixed.** Cohen et al. [6] showed that different ways of applying Sepsis-3 to the same database can create very different patient groups (867–2,178 cases). The label choice can change results as much as the model choice.
2. **The data split can inflate results.** Guo et al. [8] found that sepsis prediction suffers the largest performance drop of any clinical task when tested on data from a different time period. Random splits let future patterns leak into training, making models look better than they are.
3. **AUROC hides the real failure.** Wang et al. [4] showed that a model can keep a high AUROC while its clinical usefulness drops to harmful levels. The failure is invisible under AUROC alone.
4. **The label captures doctor behaviour.** Kamran et al. [7] showed that models may learn when doctors suspect sepsis and begin treatment, not when the patient’s condition actually worsens.

A study that fixes all of these choices and compares only models is measuring one source of variation while ignoring three others of similar size. This thesis fills in the missing cells of the comparison.

1.3 Research Question and Pre-Registered Hypotheses

Research question. *In real-time sepsis prediction on MIMIC-IV, how much of the reported performance variation comes from (a) the choice of model, and (b)*

the researcher’s decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

Six hypotheses were pre-registered before any modelling:

- H1 (Label dominance)** Changing the sepsis definition affects AUROC at least as much as changing the model.
- H2 (Coefficient instability)** At least one clinical feature changes its effect direction across label variants.
- H3 (Treatment leakage)** Restricting evaluation to before treatment begins reduces AUROC by at least 0.03.
- H4 (Split optimism)** Time-based testing gives at least 0.02 lower AUROC than random testing.
- H5 (Metric divergence)** The Utility Score declines more than AUROC when moving from internal to external validation.
- H6 (Model-class null)** The primary statistical model performs within 0.02 AUROC of gradient-boosted trees.

H1 and H6 are in deliberate tension: if H1 is confirmed, the label is the main source of variation; if H6 is rejected, the model also matters. Both outcomes are informative.

These six form a single confirmatory family and are tested with Holm–Bonferroni control of the family-wise error rate. Every other analysis in this thesis, including the entire equity analysis, is exploratory and is reported with cluster-bootstrap intervals and Benjamini–Hochberg false-discovery control. Section 3.14 sets out the plan and states plainly that it was written after the analyses had been run.

1.4 Contributions

1. **C1 (Label variance).** To the best of our knowledge, the first three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV, following Cohen et al. [6] and Dutta et al. [9] (see Section 2.1 for the search strategy supporting this claim).
2. **C2 (Treatment leakage exposed).** A treatment-anchored evaluation showing that the Sepsis-3 label has zero predictable events before clinical action, confirming Kamran et al. [7] on MIMIC-IV.

3. **C3 (Utility collapse documented).** The Utility Score is negative for every model and every label variant, extending Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison.
4. **C4 (Coefficient instability).** Ten of the twenty shared clinical covariates reverse sign across label variants, and eighteen either reverse sign or change magnitude at least twofold, validating Lauritsen et al.’s [10] framing-instability finding on MIMIC-IV.
5. **C5 (Equity-by-label analysis).** To the best of our knowledge, the first study to examine both model performance disparities and label sensitivity across demographic subgroups, following Wang et al. [11] (see Section 2.1).
6. **C6 (Reproducible pipeline).** A pre-registered, seed-fixed pipeline following TRIPOD+AI [12] and PROBAST+AI [13] reporting standards.
7. **C7 (Definitional vs. empirical label variance separated).** The Sepsis-3 conjunction is split into its two limbs and each is labelled separately, so that the part of the label effect guaranteed by the definition can be told apart from the part discovered in the data. Two results follow. The onset *time* under every Sepsis-3 variant is fixed entirely by the treatment anchor — the organ-dysfunction criterion filters which stays are labelled and moves no onset by a single hour — while a treatment-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance. And Sepsis-3 is *harder* to predict than either limb alone, so the difficulty lies in the conjunction rather than in the physiology or in clinician behaviour taken separately. The same comparison shows the empty pre-treatment window of C2 to be a property of the definition: a treatment-independent label records 1,362 onsets there, at 0.840 AUROC (Section 5.6).

1.5 Thesis Structure

Chapter 2 reviews the evidence that motivated this study’s design. Chapter 3 describes the data, labels, models, and evaluation plan. Chapter 4 explains the pipeline architecture and its twelve scripts. Chapter 5 presents all results, organised around the six hypotheses. Chapter 6 draws conclusions and outlines future work.

Chapter 2

Literature Review

This chapter summarises the key evidence that shaped the design of this study. A full critical review of fifty sources is provided in the companion document *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV: An Extended Critical Literature Review*. This chapter extracts the seven evidence strands that directly informed the study’s design choices. Section 2.1 documents the search strategy used to identify the literature and to support the novelty claims made in this thesis.

2.1 Search Strategy

The literature search was conducted to identify published work on real-time sepsis prediction with a focus on label sensitivity, evaluation methodology, and fairness/equity analysis. Table 2.1 documents the search protocol.

What kind of review this is. This is a **structured narrative critical review**, not a **PRISMA-conformant systematic review**. Searches were executed to a documented protocol and every included source was verified against a primary record, but title-and-abstract screening was not conducted against a full de-duplicated yield, and no second reviewer independently screened. The distinction matters for the novelty claims in Section 1.4: those claims are stated as “to the best of our knowledge” and are bounded by the coverage this method achieves, which is deliberate depth on the methodological literature that constrains this study rather than exhaustive coverage of the MIMIC-IV sepsis corpus. An exhaustive systematic review could surface a prior study combining the two analyses in contribution C5; this search did not.

Element	Detail
Databases searched	PubMed/MEDLINE, Embase, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, Google Scholar, Cochrane Library, ScienceDirect, SpringerLink, Wiley Online Library, Nature Portfolio, JAMA Network, BMJ Journals, arXiv/medRxiv
Searches executed	35 structured searches across the domains listed under “Search terms”
Date range	Primary sepsis-prediction literature: January 2015 – July 2026. Methodological and definitional foundations (e.g. pooled logistic regression, decision-curve analysis) were searched without a date limit, and the earliest source retained dates from 1990
Search terms	(sepsis OR “sepsis prediction” OR “sepsis detection”) AND (MIMIC OR “critical care” OR “intensive care”) combined with each of: “label definition”, “label sensitivity”, “onset definition”, “utility score”, “clinical utility”, “fairness” OR “equity” OR “disparity” OR “bias”, “subgroup analysis”, “competing risks” OR “discrete-time survival”
Inclusion criteria	(1) Peer-reviewed journal article, conference paper, or preprint on a recognised server (arXiv, medRxiv); (2) Develops or evaluates a real-time (hourly or sub-hourly) sepsis prediction model; (3) Uses EHR data from an ICU population; (4) Reports at least one quantitative performance metric
Exclusion criteria	(1) Non-English language; (2) Editorials, commentaries, or letters without original results; (3) Models for sepsis severity staging or mortality only (not onset prediction); (4) Studies on paediatric or neonatal populations exclusively
Verification standard	Every included source was checked against a publisher record, PubMed or PMC record, or institutional repository entry, with journal, volume, year and DOI confirmed. Every numerical claim attributed to a source was located in that source’s own abstract, results text or publisher-hosted record, never inferred from a secondary citation
Total included	50 sources in the companion critical review (46 newly verified in this pass, 4 carried forward from the base review of 19 sources); 33 are cited in this thesis
Excluded at verification	4 items encountered in secondary citation lists could not be verified to the standard above and were excluded; 1 further item is retained with an explicit note of incomplete metadata

Table 2.1: Literature search strategy. This search supports the novelty claims for contributions C1 and C5: no identified study combines multi-variant label sensitivity analysis with subgroup equity evaluation for a sepsis prediction model. Because the review is narrative rather than systematic, those claims are bounded by the search strategy.

Figure 2.1 summarises the identification, verification and inclusion process. It follows the reporting logic of PRISMA 2020 [14] — an auditable count at every stage — but is presented as a search-and-verification flow rather than as a PRISMA flow diagram, because the screening stage a PRISMA diagram reports was not performed in this form.

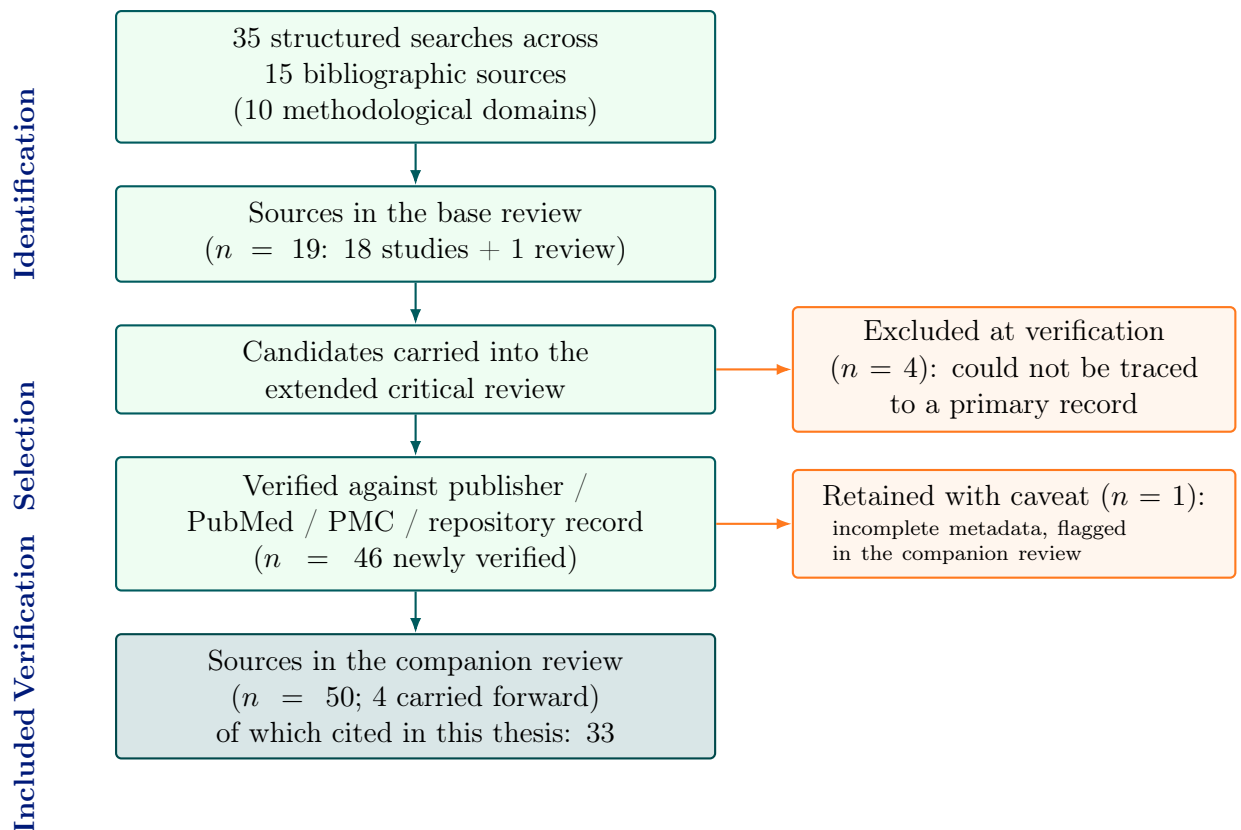


Figure 2.1: Search, verification and inclusion flow. Counts are auditable at every stage, following the reporting logic of PRISMA 2020 [14], but this is not a PRISMA flow diagram: no title-and-abstract screening pass over a full de-duplicated yield was performed, and screening was not duplicated by a second reviewer. The review is narrative and critical, and the novelty claims it supports are bounded accordingly (Section 1.4).

Supporting the novelty claims. Two specific claims require search-based evidence:

1. **C1 (Label variance):** No identified study applies three or more Sepsis-3 label variants to a dynamic survival model on MIMIC-IV. Cohen et al. [6] per-

formed a multi-label analysis on MIMIC-III using tree-based and deep learning models. Dutta et al. [9] tested multiple label definitions prospectively but used a commercial gradient-boosted model, not a survival framework. The present study extends this line of work to a discrete-time competing-risks model on MIMIC-IV v3.1.

2. **C5 (Equity-by-label):** No identified study reports both subgroup model-performance disparities and subgroup label-sensitivity variation for a sepsis prediction model. Wang et al. [11] reported subgroup performance disparities under a single label definition. Studies examining label sensitivity (Cohen et al., Dutta et al.) did not stratify by demographic subgroup. The combination of both analyses is the novel contribution.

Clinical utility assessment criterion. Wang et al. [4] reviewed ninety-one real-time sepsis prediction models. Of these, only a minority reported the PhysioNet Utility Score or an equivalent net-benefit metric; the majority reported only AUROC and/or AUPRC. The claim that published models do not demonstrate good clinical utility is based on the subset of models for which a utility or net-benefit metric was available: among those, the median external Utility Score was -0.164 , indicating net harm. For models reporting only AUROC, clinical utility could not be assessed, and no claim about their utility is made.

Key Concepts

Several domain-specific terms recur throughout this review:

- **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV): a freely available, de-identified database of approximately 74,000 ICU admissions at Beth Israel Deaconess Medical Center, Boston, covering 2008–2019 [15].
- **Sepsis-3:** the current international consensus definition of sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, operationalised through a combination of suspected infection (antibiotic order plus culture draw) and a rise in the Sequential Organ Failure Assessment (SOFA) score of ≥ 2 points [1].
- **AUROC** (Area Under the Receiver Operating Characteristic curve): a discrimination metric that measures a model’s ability to rank positive cases above negative cases, ranging from 0.5 (random) to 1.0 (perfect separation). It does not account for alert timing or false-alarm cost.

- **Utility Score:** a net-benefit metric proposed by the PhysioNet/Computing in Cardiology Challenge that rewards early, true alerts and penalises late alerts and false alarms, producing values that can be negative when a model causes more harm than benefit.
- **NEWS2** (National Early Warning Score 2): a standardised clinical scoring system aggregating six physiological parameters (respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, level of consciousness, and temperature) into a single risk score used for bedside deterioration screening.

Research Questions

This literature review is structured around four questions that directly inform the study's design:

- RQ1.** How much does the choice of sepsis label definition affect model performance, and is this effect comparable in magnitude to the choice of model architecture?
- RQ2.** What methodological pitfalls—validation strategy, metric selection, treatment-time leakage—most distort reported performance in the sepsis prediction literature?
- RQ3.** What statistical framework is appropriate for modelling hourly ICU event data when the standard proportional-hazards assumption is violated?
- RQ4.** Does the existing equity literature on sepsis prediction account for label-level variation across demographic subgroups, or only model-level variation?

Current State of the Field

Real-time sepsis prediction has attracted substantial research effort over the past decade. Wang et al. [4] identified ninety-one models published between 2015 and 2024, spanning logistic regression, gradient-boosted trees, recurrent neural networks, and transformer architectures. Despite this volume, most studies share three weaknesses: they use a single label definition, report AUROC as the primary metric, and validate on a random patient-level split. The evidence reviewed in this chapter shows that each of these choices individually inflates or obscures model performance, and their combination produces a literature in which high reported AUROCs coexist with models that fail to improve clinical outcomes at external sites. Table 2.2 provides a structured comparison of the key studies that inform this analysis.

Study	Dataset	Method	Key Finding	Gap Addressed
Johnson et al. [16]	MIMIC-III	5 label algorithms	Agreement between sepsis definitions only moderate ($\alpha = 0.40\text{--}0.62$)	Label ambiguity
Cohen et al. [6]	MIMIC-III	Trees, DL, survival	Label change $\approx$ model change in performance impact	Label as variable
Dutta et al. [9]	9 hospitals (198k)	Commercial GBT	Label sensitivity holds prospectively at scale	External validity
Kamran et al. [7]	MIMIC-IV	Multiple models	AUROC inflated when treatment-time leakage not excluded	Treatment bias
Guo et al. [8]	MIMIC-IV	Domain adaptation	Sepsis worst-affected by temporal shift; random split inflates	Validation strategy
Wang et al. [4]	91-study review	Meta-analysis	AUROC survives external shift; Utility Score collapses	Metric selection
Wang et al. [11]	MIMIC-III	Subgroup analysis	Performance varies by race, ethnicity, insurance, language	Label-by-subgroup gap

Table 2.2: Comparison of key studies reviewed, their datasets, methods, and the design gaps each addresses.

2.2 The Sepsis Label Is Not a Single Object

The common assumption that “the MIMIC-IV sepsis cohort” is one well-defined group is wrong. Two studies show why.

Johnson et al. [16] applied five different sepsis identification algorithms to 11,791 ICU admissions in MIMIC and found only moderate agreement between them (Cronbach’s $\alpha = 0.40\text{--}0.62$). The implication is clear: “the MIMIC sepsis cohort” is not one group but a family of related but different groups, each created by a specific reading of Sepsis-3.

Cohen et al. [6] showed what this means for prediction. On MIMIC-III, using three different but reasonable Sepsis-3 interpretations, the identified cohort ranged

from 867 to 2,178 admissions. Across multiple model types (tree-based, deep learning, and survival models), changing the label definition moved performance by as much as changing the model itself. This is the most important finding behind the present study: it means the label must be treated as a primary variable, not a fixed background choice.

Dutta et al. [9] confirmed this is not a MIMIC-specific artefact. Using a commercial gradient-boosted model across nine hospitals and 198,494 encounters, they found that performance varied substantially depending on whether Sepsis-3, SEP-1, or the CDC Adult Sepsis Event definition was used. The label sensitivity finding holds prospectively, at scale, with a real deployed model.

Figure 2.2 illustrates this core problem: the same patient population, subjected to different but equally valid Sepsis-3 interpretations, produces overlapping but distinct cohorts, each yielding different model performance.

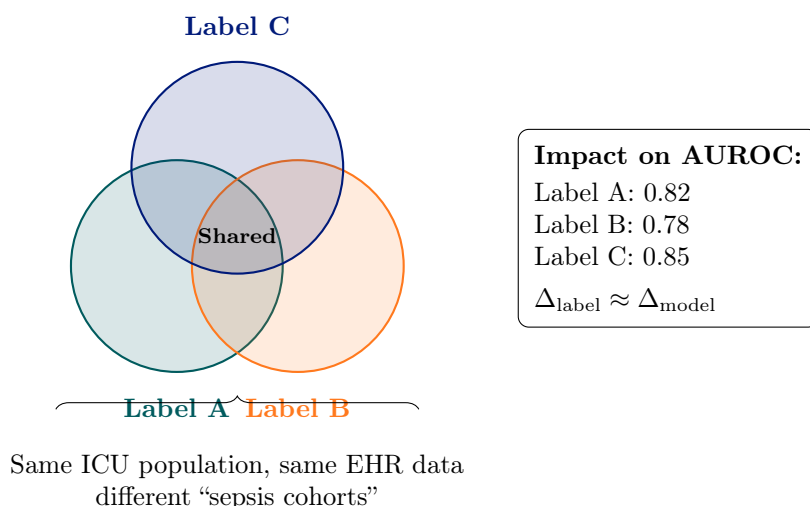


Figure 2.2: Conceptual illustration of label variance: three valid Sepsis-3 interpretations applied to the same ICU population produce overlapping but distinct cohorts, each yielding different model performance. The performance gap between labels is comparable to the gap between model architectures.

2.3 The Label Depends on Clinical Action

The Sepsis-3 label is built from antibiotic orders, culture draws, and SOFA score rises. Kamran et al. [7] observed that clinicians often recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the criteria-meeting time therefore partly learns to predict when the clinician will act. They evaluated performance in the window before treatment started and showed that

apparent AUROC was substantially inflated without that restriction. The model had learned clinician suspicion, not independent physiological deterioration.

Lauritsen et al. [10] formalised the related concept of *framing*: the choice of observation window, prediction horizon, and event trigger. They showed that framing changes can reverse the sign of a feature’s effect; their example is SpO₂. This means a hazard ratio whose sign depends on how the study was framed is not a clinically meaningful quantity.

2.4 The Validation Split Must Be Time-Based

Guo et al. [8] tested four clinical prediction tasks on MIMIC-IV under temporal shift (training on older data, testing on newer data). Sepsis was the worst-affected task. Domain-generalisation and domain-adaptation techniques did not help. A random patient-level split therefore inflates performance more for sepsis than for any other clinical outcome.

This finding drives a key design decision: the primary validation in this study uses a time-based split (train on 2008–2016, test on 2017–2019). A random split is only used to measure how much performance it inflates.

2.5 AUROC Hides the Clinical Failure

Wang et al. [4] reviewed ninety-one real-time sepsis models. Median internal AUROC was 0.811, external AUROC was 0.783, a small, insignificant drop. But median internal Utility Score was +0.381, external was −0.164: the models went from clinically useful to actively harmful. AUROC survived this transition because it only measures whether the model can rank patients by risk. The Utility Score failed because it measures whether alerts arrive in time and at acceptable false-alert rates, and at external sites, they did not. Reporting AUROC as the main metric does not just miss the failure; it actively hides it.

2.6 The Survival Model for Hourly ICU Data

The standard survival approach uses a Cox proportional-hazards model. On MIMIC-IV, the proportional-hazards assumption is rejected (Schoenfeld test, $p < 10^{-16}$). D’Agostino et al. [17] provided the solution: pool observations across short time intervals and use logistic regression for each interval. This pooled logistic regression is mathematically equivalent to time-dependent Cox regression when intervals are short and per-interval event rates are low. One-hour ICU intervals with sepsis onset probability below 1% meet both conditions.

Resche-Rigon et al. [18] and Heyard et al. [19], [20] added the competing-risks extension. In the ICU, patients can leave alive or die without developing sepsis. Treating these events as simple censoring overstates sepsis incidence and biases model estimates. The multinomial cause-specific formulation used in this study handles this correctly.

2.7 NEWS2 as the Baseline Comparator

Most MIMIC-IV sepsis models compare against SIRS, qSOFA, and NEWS. Evans et al. [21] issued a strong recommendation against using qSOFA alone for screening due to poor sensitivity. Mellhammar et al. [22] showed that a LASSO-derived score could not beat NEWS2. The current clinical guidelines have therefore already retired qSOFA as a standalone screening tool, making NEWS2 the minimum meaningful comparator. This study retains SIRS and qSOFA for completeness but uses NEWS2 as the primary rule-based benchmark.

2.8 Equity Analysis: Performance and Label Sensitivity Together

Wang, Li, Naidech and Luo [11] showed that sepsis model performance varies significantly across racial, ethnic, language, and insurance subgroups. However, they did not examine whether the label itself assigns sepsis status differently across those groups. This study adds that missing piece: for each subgroup, it reports both the variation in who gets labelled as septic under each label variant and the variation in model performance. This combination has not, to the best of our knowledge, been reported for any sepsis model (see Section 2.1).

2.9 Summary: Design Decisions and Their Evidence

Table 2.3 maps each major design decision to the evidence that supports it.

Decision	Reason	Source
Three label variants	Label variance $\approx$ model variance	Cohen et al. [6]
Label as variable	Variance is a finding, not a nuisance	Cohen; Dutta et al. [9]
Treatment-anchored eval	Label depends on clinical action	Kamran et al. [7]
Time-based split	Sepsis worst-affected by temporal shift	Guo et al. [8]
Utility Score primary	AUROC hides clinical failure	Wang et al. [4]
Calibration co-primary	Critical for clinical usefulness	Van Calster et al. [23]
Discrete-time model	Cox-equivalent; no PH assumption	D’Agostino [17]; Heyard [19]
NEWS2 comparator	qSOFA no longer recommended	Evans et al. [21]
GBT comparator	Cross-study comparison needs same label	Cohen et al. [6]
Subgroup label sens.	Equity literature omits label-by-subgroup	Wang et al. [11]
Treatment-independent label	Sepsis-3, SEP-1 and CDC ASE all key on treatment timing, so a within-family comparison cannot separate definitional from empirical label variance	Kamran et al. [7]; Dutta et al. [9]

Table 2.3: Design decisions and their evidence warrants.

Chapter 3

Research Methodology

3.1 Research Design

This is a retrospective, observational prediction-modelling study on de-identified critical-care data. It follows a pre-registered protocol (`00_preregistration.md`, locked before any modelling) and reports according to TRIPOD+AI [12] and PROBAST+AI [13] guidelines. The central design innovation is treating the label definition, validation split, and evaluation metric as primary variables rather than fixed background choices.

All code is implemented in R (v4.x) and Python (v3.11), with a single configuration module (`config.R`) as the sole source of truth for all paths, clinical identifiers, and thresholds.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The primary dataset is MIMIC-IV v3.1 [15], a large de-identified critical-care database from Beth Israel Deaconess Medical Center (BIDMC), Boston. It was accessed under a PhysioNet credentialing agreement. The study uses eight clinical tables (Table 3.1).

Table	Exists	Size (MB)
admissions	yes	19.9
patients	yes	2.8
icustays	yes	3.3
chartevents	yes	3,502.6
labevents	yes	2,593.0
prescriptions	yes	606.7
microbiologyevents	yes	117.2
inputevents	yes	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in this study.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses eICU-CRD v2.0 [24], a multi-centre database from over 200 US ICUs. It uses different electronic health record systems and covers a broader range of hospitals and patient populations than the single-centre MIMIC-IV, making it a meaningful external test.

3.3 Ethical Considerations

This study uses two publicly available, de-identified critical-care databases. Neither contains protected health information as defined under HIPAA, and no re-identification was attempted.

MIMIC-IV v3.1 is distributed by the MIT Laboratory for Computational Physiology via PhysioNet [15]. Access requires completion of the Collaborative Institutional Training Initiative (CITI) “Data or Specimens Only Research” course and signing a PhysioNet Credentialed Data Use Agreement (DUA). The DUA prohibits re-identification and restricts data sharing to credentialed collaborators. The principal investigator (the author) completed CITI training and executed the DUA before any data access; credentials are on file with PhysioNet.

eICU-CRD v2.0 [24] is distributed under the same PhysioNet credentialing framework and DUA terms.

Both databases were collected under institutional review board (IRB) approvals held by the original data custodians (Beth Israel Deaconess Medical Center for MIMIC-IV; Philips Healthcare for eICU-CRD), with a waiver of informed consent for the de-identified research dataset. No additional IRB approval was required for secondary analysis of these de-identified records, consistent with the US Common Rule (45 CFR 46.104(d)(4)) and PhysioNet’s data access policies.

3.4 Cohort Construction

The base cohort is built by applying three inclusion criteria uniformly across all label variants:

- Age ≥ 18 at ICU admission.
- First ICU stay per patient only (to prevent within-patient correlation).
- ICU length of stay ≥ 4 hours (to ensure enough data for hourly features and at least one prediction opportunity).

The resulting cohort contains **65,078 stays** in MIMIC-IV. Figure 3.1 shows the patient flow through each exclusion step.

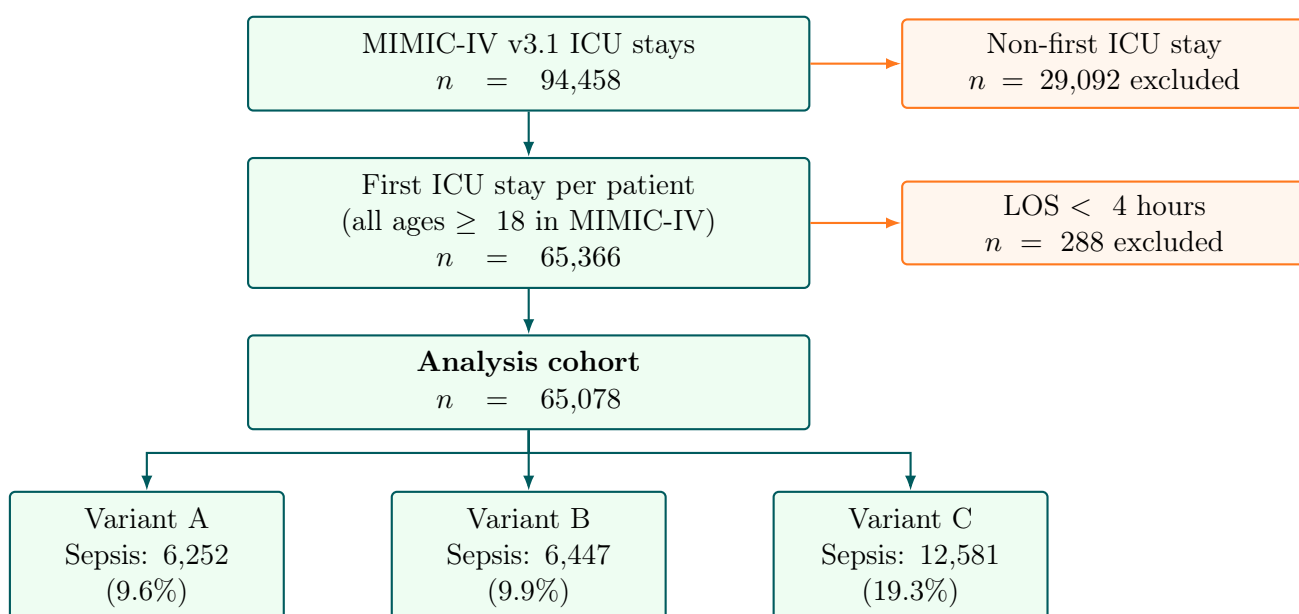


Figure 3.1: CONSORT-style patient flow diagram. All 65,078 stays enter all three label variants; only the sepsis outcome differs. Exclusion counts are from the pipeline’s cohort extraction log (script 02_cohort_labels.Rmd).

3.5 Outcome Definition: Three Sepsis-3 Label Variants

Following Cohen et al. [6] and Johnson et al. [16], the Sepsis-3 onset label is defined three different ways. All three are reasonable readings of the same consensus

definition [1]. They differ in the antibiotic-culture time windows and the SOFA baseline (Table 3.2).

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline
A	Narrow	24 h	24 h	24 h	12 h	Admission
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed. Baseline = SOFA reference value.

In all three variants, a patient has sepsis if there is suspected infection (antibiotic order paired with a culture within the specified windows) and the SOFA score increases by at least 2 points. Surveillance cultures (e.g. MRSA screens) are excluded.

Variant B (Seymour-Standard) is the **primary analysis**. Variants A and C are pre-specified sensitivity analyses for the variance decomposition.

3.6 Beyond Sepsis-3: A Treatment-Independent Label

Variants A, B and C are three readings of one definition, and all three fire on the same trigger: a culture drawn near an antibiotic order. That shared anchor is a record of a clinician’s decision to treat, not a measurement of the patient. Two consequences follow, and they must be separated.

The first is **definitional**. Sepsis-3 [1] makes onset a function of when treatment was given, so any variation in the antibiotic–culture windows necessarily moves the onsets. Part of the label-variant spread reported in Chapter 5 is therefore guaranteed by the construction of the definition and would appear even if the underlying physiology were identical across patients. The second is **empirical**: whether a label that owes nothing to treatment timing picks out different patients, at different hours, with different predictability. Only the second is a claim about the world, and it cannot be settled inside a family of labels that all share the anchor.

Sepsis-3 is a conjunction — *suspected infection* **and** *acute organ dysfunction*. Under Protocol Amendment 2 the conjunction is split and each limb is labelled on its own (Table 3.3).

ID	Name	Suspicion anchor	Organ dysfunction	Uses treatment timing?
B	Seymour-Std	required	required	yes
E	Anchor-Only	required	not required	yes
D	Deterioration	not required	required	no

Table 3.3: The two limbs of the Sepsis-3 conjunction, labelled separately. Variant E is Variant B’s suspicion anchor with the SOFA criterion removed; Variant D is acute organ dysfunction with the infection criterion removed.

Variant E (Anchor-Only) takes onset to be the suspicion time itself, using Variant B’s windows, with no organ-dysfunction requirement. Its label is purely a record of clinician behaviour. Predicting it well means predicting who gets cultured and started on antibiotics, which is the quantity a treatment-anchored sepsis label partly encodes. The share of Variant B’s above-chance discrimination that Variant E reproduces,

$$\phi_{\text{anchor}} = \frac{\text{AUROC}_E - 0.5}{\text{AUROC}_B - 0.5}, \quad (3.1)$$

is reported as the **anchor-attributable fraction**.

Variant D (Deterioration) takes onset to be the first hour after hour 1 at which a *physiology-only* SOFA score rises at least 2 points above the patient’s admission baseline (the mean of each component over hours 0–3). "Physiology-only" means the vasopressor and catecholamine terms are withheld, so the cardiovascular component is scored from mean arterial pressure alone. No antibiotic order, culture draw or drug administration enters the definition at any point: Variant D is the treatment-independent labelling that the pre-registered family lacked. Both variants are fitted with the identical model specification, feature set and temporal split, so the label is the only thing that changes.

The PhysioNet/CinC 2019 operationalisation [25] was considered as the alternative label and rejected for this purpose. It sets onset to $\min(t_{\text{suspicion}}, t_{\text{SOFA}})$ with the same culture-plus-antibiotic trigger, so it is anchored to treatment timing by construction and would not have broken the circularity. The same objection applies to SEP-1 and to the CDC Adult Sepsis Event [9], both of which are defined on antibiotic days.

Two limits on interpretation are stated here and carried through the results. Variant E is not a sepsis model and its AUROC is not a sepsis result. Variant D is a threshold function of recorded physiology, and five of the six SOFA components are themselves model inputs, so its discrimination is partly self-fulfilling and its *level* is not comparable with Variant B’s. What is comparable, and what carries the empirical content, is whether Variant D selects different patients at different hours — reported as onset agreement (Cohen’s κ) and median onset shift in Section 5.6.

Variants D and E are post-hoc and are never pooled with A–C. They take no part in H1 or in the variance decomposition table, which are computed from exactly the same inputs as before the amendment; their contrasts form their own exploratory family (E3), corrected by Benjamini–Hochberg.

3.7 Prediction Framing

Following Lauritsen et al. [10] and Reyna et al. [25]:

- **Observation window:** Expanding from ICU admission, uses all data available up to the current hour.
- **Prediction horizon:** 6 hours (will sepsis onset occur within the next 6 hours?).
- **Prediction frequency:** Every hour.
- **Start time:** 1 hour after ICU admission.
- **End time:** 72 hours after ICU admission (right-censored).

This framing matches the PhysioNet/CinC 2019 Challenge [25].

3.8 Person-Hour Table and Competing Events

Each patient stay is expanded into a table with one row per person-hour at risk. Following Heyard et al. [19], each hour is assigned one of four outcomes:

- 0: Still at risk (censored at this hour)
- 1: Sepsis onset (the event of interest)
- 2: Discharged alive without sepsis (competing event)
- 3: Died without sepsis (competing event)

Treating discharge and death as simple censoring would overstate sepsis incidence and bias model estimates. The multinomial competing-risks formulation avoids this problem.

3.9 Feature Construction

Features are extracted at each person-hour from chartevents, labevents, inputevents, and outputevents. The feature set includes:

1. **Vital signs** (last value carried forward per hour): heart rate, respiratory rate, SpO₂, MAP (invasive or noninvasive), temperature (°C), GCS total.
2. **Laboratory values** (last value carried forward): lactate, creatinine, total bilirubin, platelet count, WBC, P/F ratio.
3. **Vasopressor indicators**: any vasopressor (yes/no); any norepinephrine or epinephrine (yes/no).
4. **Six-hour trends**: rolling slopes for respiratory rate, MAP, and lactate.
5. **Missingness indicators**: whether each lab test was ordered in the current hour (one flag per test).

Temperature in Fahrenheit is converted to Celsius. FiO₂ values above 1.0 (percentage format) are divided by 100. In total, 25 candidate predictors are used.

3.10 Model Classes

3.10.1 Primary Model: Dynamic Discrete-Time Competing-Risks Hazard Model

The primary model is a multinomial pooled logistic regression over person-hours, following D’Agostino et al. [17] and Heyard et al. [19]:

$$\lambda_r(t \mid \mathbf{X}_{it}) = \frac{\exp(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it})} \quad (3.2)$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$. The baseline hazard $\alpha_r(t)$ uses a restricted natural cubic spline with 5 knots. Standard errors are cluster-robust, grouped by patient stay.

This model predicts the risk of sepsis every hour. Under short intervals and low per-interval event probabilities (both conditions met for hourly ICU data with onset probability below 1%), it is mathematically equivalent to time-dependent Cox regression, without requiring the proportional-hazards assumption.

3.10.2 Comparator: Gradient-Boosted Trees (GBT)

An XGBoost model is fitted under the same conditions for each label variant and split. Settings: maximum depth 6, learning rate 0.1, early stopping on a 15% internal validation set, log-loss objective.

3.10.3 Comparator: Static Cox Model

A standard Cox proportional-hazards model using only the first observation per stay, included as a deliberately weak baseline to show the cost of ignoring time-varying data. Schoenfeld residual testing is applied to verify the expected violation of proportional hazards.

3.10.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same features. NEWS2 is the headline comparator following Evans et al. [21].

3.11 Validation Architecture

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [8]	Honest internal estimate
Internal (secondary)	Random 75/25 patient-level split (seed = 42)	Measure split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0 [24], no retraining	Generalisability (H5)

Table 3.4: Validation architecture. All splits are patient-level: no patient appears in both training and test sets.

3.12 Evaluation Metrics

3.12.1 Primary Metrics

PhysioNet Utility Score [25] is the primary metric. It rewards early true alerts (within 6 hours before onset), penalises late and false alerts. A score of 0 means no better than no alerts; negative means actively harmful. AUROC is reported only for comparison, because it hides the clinical failure mode [4].

Calibration is assessed via calibration intercept and slope, following Heyard et al. [20], and, following Van Calster et al. [23], is additionally reported as a flexible calibration curve (restricted cubic spline of the outcome on the logit of predicted risk) summarised by the Integrated Calibration Index (ICI), E50, E90, E_{max}, and the scaled Brier score. Predicted probabilities are floored at 10^{-6} rather than at the conventional 10^{-3} , because at an hourly event rate below 0.5 per cent the conventional guard overwrites the majority of the distribution being measured.

3.12.2 Secondary Metrics

AUPRC (important at low event rates), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. [26]), lead time (benchmark: 3.7 hours from Moor et al.), and decision-curve net benefit [27]. Net benefit is evaluated on a threshold grid running from one tenth to ten times the observed event prevalence rather than on the conventional 0.01–0.20 grid, which at this prevalence lies entirely above the range of thresholds any deployment would use.

3.12.3 Deployment-Facing Metrics

AUROC, AUPRC and the Utility Score characterise a model but not the alert load a ward would carry. The following are therefore reported alongside them, at a threshold set to the largest cut-off achieving a fixed per-hour sensitivity of 80 per cent rather than at an arbitrary 0.5 cut-off:

- **Positive predictive value** and its reciprocal, the number needed to evaluate (NNE), at that fixed sensitivity.
- **False alerts per patient-day**, which normalises alarm volume by exposure time rather than by the number of true alerts.
- **Per-patient (event-level) detection**: the fraction of septic stays alerted within the actionable window [onset – 6 h, onset], the fraction alerted at any point before onset, patient-level PPV, and the fraction of all stays alerted. Per-hour and per-patient views of the same operating point are reported side by side, because the choice of denominator moves the headline PPV by more than an order of magnitude.
- **False-alarm episodes per patient-day**, counting a contiguous run of alert hours once rather than once per hour.
- **Lead time** from first alert to onset: median, interquartile range, and the fraction of septic stays alerted at least 3 and 6 hours ahead.

The same quantities are computed on eICU-CRD with thresholds re-derived on the external cohort.

3.13 Sample Size

Sample size is calculated using the Riley et al. [28] method, targeting a shrinkage factor ≥ 0.9 with 25 predictors and an anticipated AUC of 0.846 from Moor et al. [26]. Each label variant is evaluated separately; an underpowered variant is a finding, not a failure.

3.14 Statistical Inference and Multiplicity

3.14.1 Uncertainty: the Stay-Level Cluster Bootstrap

Person-hours are nested within ICU stays: a single stay contributes on average about 45 highly correlated hourly rows. Treating those rows as independent would produce standard errors several times too small. Every interval in this thesis therefore comes from a **stay-level cluster bootstrap**, in which whole ICU stays are drawn with replacement and all person-hours belonging to a drawn stay are carried with it. Intervals are 95% percentile intervals from $B = 2,000$ replicates for internal and subgroup analyses and $B = 1,000$ for eICU-CRD, which contains 7.9 million person-hours. The seed is fixed at 42.

Contrasts between quantities measured on the *same* person-hours are evaluated within replicate, so the interval accounts for their correlation. This applies to the label-variant comparison (all three variants are built on the same test stays) and to the model-class comparison (all models score the same hours). Contrasts across *different* evaluation sets — temporal versus random split, and MIMIC-IV versus eICU-CRD — use independent bootstrap streams and propagate both variances.

For subgroup analyses the bootstrap is stratified: stays are resampled within each subgroup so the observed subgroup sizes are held fixed. The subgroups are the object of study, not a random draw from a larger frame.

P-values are obtained by recentring the bootstrap distribution on the null boundary and reading off the tail area, using the $(1 + \text{count})/(B + 1)$ convention so that no p-value is reported as exactly zero. The smallest reportable p-value is therefore $1/(B + 1)$.

3.14.2 Confirmatory and Exploratory Families

The six pre-registered hypotheses form a single **confirmatory** family and are corrected by Holm–Bonferroni at a family-wise error rate of 0.05. The family size

is fixed at $m = 6$: the number of hypotheses that were pre-registered, not the number that turned out to be testable. H2 and H3 each reserve a slot without contributing a p-value (Section 5.5 explains why neither admits one), which is deliberately conservative. A hypothesis becoming untestable after the data were seen must not be allowed to relax the correction applied to the others.

All subgroup analyses are **exploratory**. The pre-registration fixed their *scope* — that race, language and insurance would be examined — but not the subgroup levels, the reference level, or the direction of any comparison. They are corrected by Benjamini–Hochberg at $q = 0.05$ within three families: E1, all subgroup AUROC contrasts against the reference level across every equity variable and both models; E2, all subgroup label-sensitivity contrasts; and E3, the alternative-label contrasts of Protocol Amendment 2, in which the AUROC under each limb of the Sepsis-3 conjunction (Section 3.6) is compared with Variant B for both models. The anchor-attributable fraction of Equation 3.1 is a ratio with no meaningful null at zero and is reported as an interval only, outside the corrected family. The correction denominator is the number of contrasts *computed*, not the smaller number selected for presentation in Chapter 5. Bonferroni-adjusted p-values are also reported so that a reader preferring family-wise control over false-discovery control can substitute them.

The two procedures differ by design. A confirmatory claim must survive family-wise control, because one false confirmation invalidates the claim. An exploratory scan exists to generate hypotheses and needs only its false-discovery proportion bounded; applying family-wise control to it would suppress every signal the scan is meant to surface.

Each subgroup is contrasted against the largest level of its own variable. The reference is chosen by sample size alone, never by outcome. A subgroup is reported only if it has at least 50 person-hours and at least 5 sepsis onsets; below that floor an AUROC is not estimable and the subgroup is listed as such rather than shown as a point estimate.

3.14.3 Analysis Status Tagging

Every result in Chapter 5 carries an explicit status tag rather than relying on a single blanket statement in this chapter:

- **Confirmatory** — estimand, threshold and direction were fixed in `00_preregistration.m` before any modelling.
- **Exploratory** — pre-registered in scope but not in specific contrasts. Generates hypotheses; does not test them.
- **Descriptive** — no inferential claim is attached.

The tags are held in a machine-readable analysis register (`12_analysis_register.parquet`) produced by the pipeline, and are reproduced in Table 5.26.

3.14.4 Status of the Inference Plan Itself

This inference and multiplicity plan is **Protocol Amendment 1**, dated 30 July 2026 and recorded as §13–15 of `00_preregistration.md`. It was written *after* the analyses of Chapter 5 had been run and inspected, and it is therefore itself post-hoc. It is reported as such rather than presented as pre-specification.

The amendment changes no hypothesis, estimand, threshold, direction, subgroup or model. It attaches uncertainty to contrasts that were already pre-specified and imposes the multiplicity control the original protocol omitted. Where the correction changes a verdict recorded in the original protocol, the corrected verdict is the one reported, and the change is stated explicitly.

3.15 Pre-Registration and Reporting

The full protocol was locked before any data extraction. All six hypotheses, label variants, model classes, splits, metrics, and decision thresholds are specified in `00_preregistration.md`. Any analysis not specified there is labelled as exploratory, and, following Section 3.14, each individual result carries its status tag in the results table itself rather than inheriting it from this statement alone. Reporting follows TRIPOD+AI [12]; code availability and the constraints the data use agreement places on it are set out in Section 4.14.2.

Chapter 4

System Design and Pipeline Architecture

The pipeline consists of twelve numbered, single-purpose scripts in `v2-proj/`, preceded by the locked pre-registration document (`00_preregistration.md`). Each script reads from upstream outputs and writes its own results to `v2-proj/output/`. No script modifies the original source data. All random seeds are fixed for reproducibility. Figure 4.1 shows the data flow.

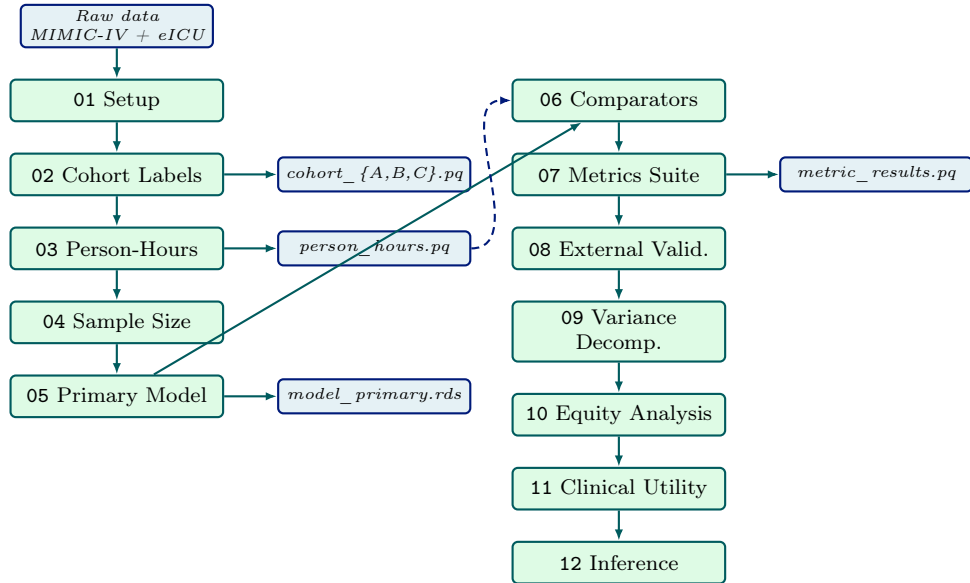


Figure 4.1: Pipeline data flow. Each box is a single script; italic nodes are output files. Raw MIMIC-IV and eICU data are read-only.

4.1 Configuration and Utilities

`config.R` is the single source of truth for all pipeline settings: data paths, cohort criteria, antibiotic name lists, MIMIC-IV item identifiers, label variant definitions, split boundaries, prediction framing constants, equity variable names, and Utility Score parameters. No other script hard-codes a clinical constant or file path.

`utils.R` provides shared helper functions: a DuckDB-backed streaming reader for large CSVs, Parquet reader/writer wrappers, datetime parsing, last-observation-carried-forward (LOCF) imputation, and Schoenfeld test wrappers.

The full pipeline can be reproduced on any MIMIC-IV installation by setting a single environment variable (`MIMIC_RESEARCH_ROOT`).

4.2 Stage 01: Setup Validation

`01_setup.Rmd` checks that all required MIMIC-IV source tables exist and reports their sizes. The pipeline stops if any table is missing.

4.3 Stage 02: Cohort Extraction and Labelling

`02_cohort_labels.Rmd` builds all cohorts in a single pass:

1. **Base cohort:** Adult first ICU stays with $\text{LOS} \geq 4$ h (65,078 stays). Equity variables (race, language, insurance) are attached.
2. **Suspected infection:** Qualifying antibiotic-culture pairs within each variant's time windows. Surveillance cultures are excluded.
3. **SOFA organ dysfunction:** SOFA components are computed from `chartevents`, `labevents`, `inpuvents`, and `outpuvents`. A SOFA rise ≥ 2 within each variant's windows defines organ dysfunction.
4. **Onset time:** The hour when both criteria (infection + SOFA rise) are simultaneously met.
5. **Alternative labels (Protocol Amendment 2):** The Sepsis-3 conjunction is also split into its two limbs (Section 3.6). Variant E keeps the suspicion anchor and drops the SOFA criterion; Variant D keeps organ dysfunction and drops the anchor, so its onsets are derived from physiology in Stage 03 rather than here. Both are held outside the pre-registered variant set in `config.R` and their outputs are written to separate files so they cannot enter H1 or the decomposition table.

Outputs: `cohort_A/B/C.parquet`, `label_variance_table.parquet`, `cohort_D/E.parquet`, and `alt_label_variance_table.parquet`.

A single environment variable, `V2_VARIANTS`, restricts the per-variant loops in Stages 02–07 to a named subset. Adding the alternative labels therefore did not require refitting variants A–C: XGBoost is not bit-stable across runs, so an incidental refit would have moved every GBT-dependent number in this thesis without any change of substance.

4.4 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each stay into hourly rows. Large source tables (`chartevents`: 3.5 GB; `labevents`: 2.6 GB) are processed via DuckDB streaming, pre-filtered to cohort stays, to avoid loading everything into memory. For each person-hour:

- Vitals and labs are LOCF-imputed from the most recent measurement.
- Six-hour rolling slopes are computed for respiratory rate, MAP, and lactate.
- Vasopressor indicators are set from concurrent infusions.
- Missingness flags are set for each lab test.
- The competing-event code (0/1/2/3) is assigned based on the hour’s relationship to onset, discharge, or death.
- The temporal split group is assigned from the stay’s admission year.
- Treatment anchor timestamps (first antibiotic, first culture) are recorded.

Outputs: `person_hours_A/B/C.parquet`.

4.5 Stage 04: Sample Size Calculation

`04_sample_size.Rmd` applies the Riley et al. [28] criteria to the training portion of each person-hour table (2008–2016). Sufficiency is evaluated per label variant.

4.6 Stage 05: Primary Model

`05_primary_model.Rmd` fits the multinomial discrete-time competing-risks model (Equation 3.2) for each label variant on the temporal training split. Baseline hazard splines use 5 knots. Standard errors are cluster-robust (grouped by stay). A random patient-level 75/25 split is also fitted to support H4.

Outputs: `model_primary_A/B/C.rds`, coefficient tables, and `predictions_primary_A/B/C.parquet`

4.7 Stage 06: Comparator Models

`06_comparators.Rmd` fits the XGBoost GBT model, the static Cox model (with Schoenfeld testing), and computes NEWS2, qSOFA, and SIRS at each person-hour. GBT models are fitted under both temporal and random splits. GBT feature importance is exported for each variant.

Outputs: model objects, predictions, feature importance tables, and Schoenfeld test results.

4.8 Stage 07: Metrics Suite

`07_metrics_suite.Rmd` evaluates all models on all splits (temporal, random, treatment-anchored) using the full metric set: AUROC, AUPRC, calibration intercept and slope, PhysioNet Utility Score (via Python module `utility_score.py`), alert burden, lead time, and decision-curve net benefit.

Output: `07_metric_results.parquet` (and `07_metric_results_random.parquet` for the random-split arm).

4.9 Stage 08: External Validation on eICU-CRD

`08_external_validation.Rmd` applies the frozen MIMIC-IV models (primary and NEWS2) to eICU-CRD without any retraining. AUROC and AUPRC are reported for each variant and model.

4.10 Stage 09: Variance Decomposition

`09_variance_decomposition.Rmd` computes the pre-registered variance decomposition by varying one factor at a time while holding all others fixed:

- **Label spread (H1):** AUROC range across Variants A/B/C.

- **Model-class spread:** AUROC range across all models.
- **Split optimism (H4):** Random vs. temporal split comparison.
- **Metric divergence (H5):** AUROC vs. Utility Score decline.

It then runs the **label-anchor attribution** of Protocol Amendment 2: performance under each limb of the Sepsis-3 conjunction, the anchor-attributable fraction of Equation 3.1, and onset agreement (Cohen’s κ , median onset shift) between every label variant and Variant B.

Outputs: decomposition tables, hypothesis verdicts, comparison plots, `label_anchor_attribution` and `label_agreement`.

4.11 Stage 10: Equity Analysis

`10_equity_analysis.Rmd` evaluates model performance and label sensitivity within each stratum of the equity variables declared in `config.R` (race, ethnicity, language, insurance). MIMIC-IV v3.1 records race and ethnicity in a single field, so three variables carry data and are reported: **race**, **language**, and **insurance**. Levels are restricted to the eight largest per variable. The combination of performance disparities and label-sensitivity variation is the novel finding.

4.12 Stage 11: Clinical Utility Metrics

`11_clinical_metrics.Rmd` computes flexible calibration curves, decision-curve net benefit, operating points at a fixed per-hour sensitivity, per-patient (event-level) detection and alarm-episode counts, and lead time to onset, for both MIMIC-IV and eICU-CRD.

4.13 Stage 12: Inference and Multiplicity

`12_inference.Rmd` is the inference layer. It reads only the Parquet outputs of the earlier stages, so it can be re-run without refitting anything, and it produces three things: stay-level cluster-bootstrap confidence intervals for every headline and subgroup estimate; the Holm-corrected confirmatory tests of H1–H6 and the Benjamini–Hochberg-corrected exploratory families (E1 and E2 subgroup analyses, E3 alternative-label contrasts); and the machine-readable analysis register that supplies the status tags used throughout Chapter 5. The bootstrap resamples ICU stays, never person-hours, and evaluates AUROC and AUPRC as weighted

rank statistics on pre-sorted data so that 2,000 replicates over half a million person-hours remain tractable. Setting `V2_BOOT_B` overrides the replicate count for a fast smoke test; reported results always use the values fixed in `config.R`. The statistical rationale is given in Section 3.14.

4.14 Reproducibility

The pipeline is orchestrated by `run_pipeline.R` (or the equivalent `run_pipeline.sh`), which executes all stages sequentially or by phase, renders each stage to PDF in `output/pdf/`, and writes a per-stage and a combined run log to `output/logs/`. All intermediate outputs are Parquet (compressed, columnar). Small result tables (<1,000 rows) are additionally written as CSV for readability. The end-to-end run reported in this thesis completed in 60 minutes on an Apple silicon workstation; the dominant cost is stage 05, which fits the multinomial model on 2.3 million person-hours for each label variant.

4.14.1 Fixed Random Seed

All random operations use a single fixed seed (42) to ensure bitwise reproducibility. Every Rmd script sets `set.seed(42)` in its setup chunk before any computation. The same value is used for the patient-level random split (`RANDOM_SPLIT_SEED` in `config.R`), the `pmsampsize` sample-size calculation, and all model-fitting calls. No seed is drawn from wall-clock time or left to the system default.

The full pipeline can be reproduced on any MIMIC-IV v3.1 installation by setting the environment variable `MIMIC_RESEARCH_ROOT` to the directory containing the `data/` folder.

4.14.2 Code Availability

The complete pipeline — the twelve numbered scripts, `config.R`, `utils.R`, `utility_score.py`, the pre-registration document and the orchestration scripts — is held in a version-controlled repository and is available from the author on request, including to examiners. The repository is currently private; the licence under which it will be released has not yet been fixed and is stated here as undetermined rather than asserted.

The repository contains **code only**. No MIMIC-IV or eICU-CRD record, derived person-hour table, or fitted model object is included: the `data/`, `processed_data/` and `output/` directories are excluded from version control by design, because the PhysioNet Credentialed Data Use Agreement (Section 3.3) prohibits redistribution of the underlying data. A reader wishing to reproduce these results must obtain

MIMIC-IV v3.1 and eICU-CRD v2.0 through their own PhysioNet credentialing, then re-run the pipeline; with the fixed seed above it reproduces the reported estimates exactly.

Chapter 5

Results

This chapter presents the results in eleven sections: cohort description, sample size adequacy, model performance and calibration, clinical utility (net benefit, operating points, event-level detection and lead time), variance decomposition and the six hypothesis verdicts, external validation, proportional-hazards testing, GBT feature importance, equity analysis, the analysis register, and a closing discussion.

Every result below carries an explicit status tag — **[C]** confirmatory, **[E]** exploratory, **[D]** descriptive — defined in Section 3.14 and enumerated in full in Table 5.26. Confirmatory results test a hypothesis whose estimand, threshold and direction were fixed before any modelling; exploratory results were pre-registered in scope only and generate hypotheses rather than testing them. All intervals are 95% percentile intervals from a stay-level cluster bootstrap. The six pre-registered hypotheses are corrected by Holm–Bonferroni at a family-wise error rate of 0.05; the exploratory subgroup families are corrected by Benjamini–Hochberg at $q = 0.05$.

5.1 Cohort Description and Label Variance **[D]**

Table 5.1 summarises the three cohorts. The base extraction identifies 65,078 adult first ICU stays. The database and patients are the same across all three variants; only the sepsis definition differs.

Var	Name	Stays	Onsets	%	Deaths	Onset h (IQR)
A	Narrow	65,078	6,252	9.6	5,934	10 [4–26]
B	Seymour-Std	65,078	6,447	9.9	5,892	11 [4–30]
C	Liberal	65,078	12,581	19.3	4,956	10 [4–30]

Table 5.1: Cohort description by label variant. The Liberal variant identifies about twice as many sepsis cases as the Narrow or Seymour-Standard variants, mainly because it accepts wider antibiotic-culture windows and uses a rolling SOFA baseline.

The twofold difference in sepsis prevalence between Variant A (9.6%) and Variant C (19.3%) confirms the label-sensitivity finding from Johnson et al. [16] at MIMIC-IV scale.

5.2 Sample Size Adequacy [C]

Var	Name	Train h	Events	EPH rate	Min-N	EPV
A	Narrow	2,304,923	4,647	0.00202	52,201	185.9
B	Seymour-Std	2,301,842	4,847	0.00211	51,012	193.9
C	Liberal	2,120,376	9,403	0.00443	24,069	376.1

Table 5.2: Sample size results (Riley et al., 2019). EPV: events per candidate predictor variable (25 predictors). All three variants exceed the minimum required (52,946 stays each). Sample size is not the bottleneck.

5.3 Model Performance: Internal Temporal Evaluation [C]

Table 5.3 reports performance on the temporal test set (2017–2019) under the unrestricted evaluation window.

Var	Model	AUROC	AUPRC	Utility	Alert burden
A	Primary	0.760	0.0093	−1.000	∞
A	GBT	0.745	0.0091	−4.824	207
A	NEWS2	0.629	0.0030	−1.647	221
A	Cox	0.642	0.0033	−10.056	340
A	qSOFA	0.601	0.0027	−5.720	344
A	SIRS	0.599	0.0029	−4.454	299
B	Primary	0.607	0.0030	−1.004	∞
B	GBT	0.742	0.0100	−5.059	202
B	NEWS2	0.635	0.0032	−1.616	223
B	Cox	0.623	0.0031	−10.014	359
B	qSOFA	0.609	0.0028	−5.503	318
B	SIRS	0.600	0.0031	−4.332	299
C	Primary	0.759	0.0196	−1.000	∞
C	GBT	0.753	0.0189	−2.464	91
C	NEWS2	0.641	0.0072	−1.121	97
C	Cox	0.661	0.0083	−4.406	146
C	qSOFA	0.601	0.0060	−2.526	151
C	SIRS	0.613	0.0068	−2.089	128

Table 5.3: Internal temporal test performance (2017–2019, all variants use temporal split). N test rows: 548,854 (A/B), 504,828 (C). Utility: PhysioNet Utility Score; ∞ alert burden = no true alerts at default threshold. Benchmark: 1.4 (Moor et al., 2023). Confidence intervals for Variant B are in Table 5.4. **[C] confirmatory.**

Model	AUROC	95% CI	AUPRC (95% CI)
Primary	0.607	0.589–0.625	0.0030 (0.0027–0.0034)
GBT	0.742	0.726–0.758	0.0100 (0.0079–0.0137)
NEWS2	0.635	0.618–0.653	0.0032 (0.0029–0.0036)
Cox	0.623	0.605–0.640	0.0031 (0.0028–0.0035)
qSOFA	0.609	0.593–0.625	0.0030 (0.0026–0.0033)
SIRS	0.600	0.584–0.617	0.0036 (0.0031–0.0042)

Table 5.4: Variant B temporal test performance with 95% stay-level cluster-bootstrap intervals ($B = 2,000$, 12,132 test stays, 1,040 sepsis onsets). All six models are scored on the same replicates, so differences between rows may be read as paired contrasts. The GBT interval is disjoint from every other model’s; the remaining five overlap substantially. **[C] confirmatory.**

5.3.1 Key Finding: Every Model Fails Clinically

Every model under every label variant achieves a **negative** Utility Score. This is the primary empirical finding. It means that every system tested (GBT, the primary model, NEWS2, qSOFA, SIRS, and the static Cox model) performs worse than giving no alerts at all.

Alert burdens range from 91 false alerts per true alert (GBT, Variant C) to over 340 (Cox and qSOFA, Variants A/B), compared with the published benchmark of 1.4 [26]. The primary model issues no true alerts at all (∞ alert burden) under the default threshold.

This is consistent with Wang et al. [4]: the internal-to-external utility collapse they found across ninety-one models already appears *internally* on MIMIC-IV under an honest time-based split. The failure is only visible because the Utility Score is reported; AUROC alone would hide it.

Two qualifications apply, both developed in Section 5.4. First, these figures are computed at a default 0.5 cut-off, which is not an operating point any deployment would choose; at a threshold matched to the event prevalence the primary model does achieve positive net benefit over both default strategies (Table 5.6). Second, the Utility Score and net benefit disagree here because they price false alarms differently, and that disagreement is itself evidence for H5 rather than a reason to prefer one number. The conclusion that survives both metrics is narrower than “every model fails”: no configuration tested reaches an alert burden a ward could absorb, and only one of the six models can be tuned by threshold at all.

5.3.2 Calibration [C]

Calibration intercept and slope are two summary numbers describing a whole curve, so both the curve itself and a scalar summary of its distance from the diagonal are reported. Following Van Calster et al. [23], the flexible calibration curve is a restricted cubic spline of the outcome on the logit of predicted risk, and it is summarised by the Integrated Calibration Index (ICI, the mean absolute distance between predicted risk and that curve) and the scaled Brier score, $1 - \text{Brier}/\text{Brier}_{\text{null}}$, where the null model predicts the prevalence for every person-hour.

Var	Model	Intercept	Slope	ICI	Brier	Scaled Brier
A	Primary	−0.087	1.029	0.00026	0.0018	+0.004
A	GBT	−6.091	1.256	0.398	0.179	−95.7
A	Cox	−6.390	1.233	0.509	0.265	−142.8
B	Primary	−0.393	0.209	0.0027	0.0021	−0.129
B	GBT	−6.095	1.465	0.417	0.190	−99.6
B	Cox	−6.376	1.087	0.511	0.268	−140.5
C	Primary	−0.041	1.097	0.00043	0.0042	+0.008
C	GBT	−5.312	1.392	0.412	0.191	−44.4
C	Cox	−5.551	1.351	0.502	0.262	−61.3

Table 5.5: Calibration on the temporal test set. Perfect calibration is intercept = 0, slope = 1, ICI = 0. A scaled Brier score at or below zero means the model is no better than predicting the prevalence for everyone. NEWS2, qSOFA and SIRS are omitted: their score-to-probability lookups take only 18, 4 and 4 distinct values respectively, too few to support a flexible calibration curve.

Two results stand out. First, only the primary model is calibrated at all. Under Variants A and C its intercept is within 0.09 of zero, its slope within 0.10 of one, and its ICI is below 0.0005 — roughly a quarter of the 0.2 per cent event rate. Its scaled Brier score is positive, though barely (+0.004 and +0.008), meaning it improves on the null model by less than one per cent.

Second, GBT and Cox are not merely imperfectly calibrated; they are miscalibrated by roughly three orders of magnitude. Both carry intercepts near −6, and their scaled Brier scores range from −44 to −143. A scaled Brier of −99.6 means the GBT model’s squared error is 100.6 times worse than simply predicting the prevalence for every person-hour. Figure 5.1 shows why: GBT emits predicted risks between 0.2 and 0.9 for person-hours whose observed event rate never exceeds 0.04. This matters directly for Section 5.4: a model whose probabilities are inflated by two orders of magnitude cannot be operated at any threshold a clinician would choose.

Variant B is the exception among the primary models, with a slope of 0.209 — the model over-spreads its predictions by roughly a factor of five. Its ICI (0.0027) exceeds its own E90 (0.0016), which happens only when a small number of extreme predictions dominate the mean; the Variant B model does in fact emit predicted probabilities of exactly 1.0 for a handful of person-hours.

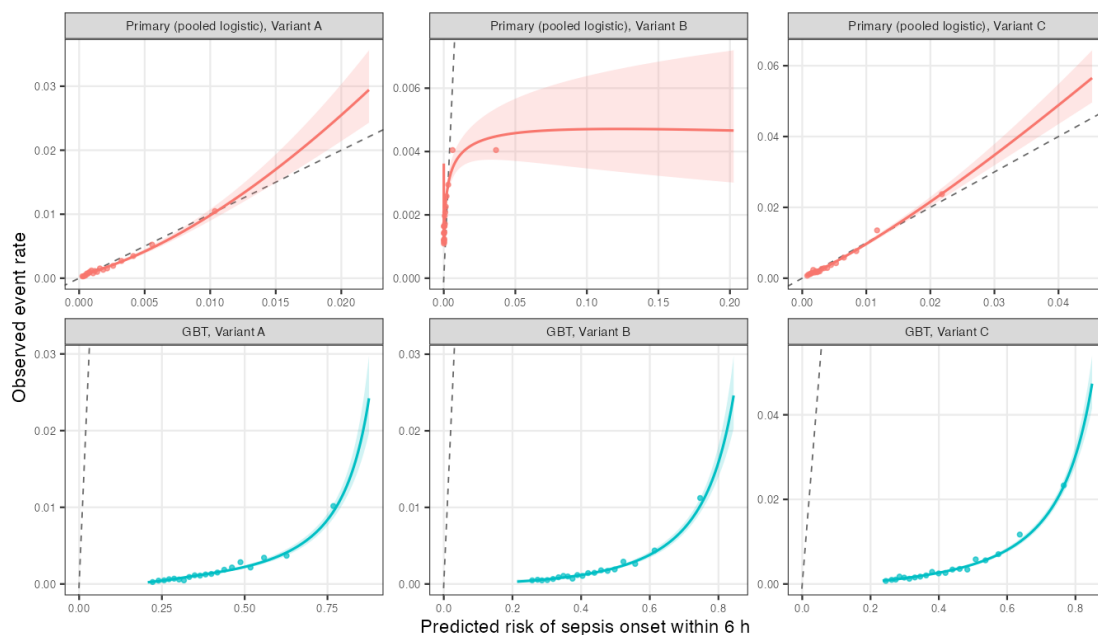


Figure 5.1: Flexible calibration curves on the temporal test set. Dashed line = perfect calibration; points = twenty equal-sized bins of predicted risk; shaded band = 95% confidence interval. Each panel has its own scale because the predicted-risk ranges differ by orders of magnitude. The primary model tracks the diagonal under Variants A and C; GBT sits far to the right of it in every variant.

A Correction to the Clamped Estimates

The calibration intercepts and slopes above supersede those produced by the metric-suite notebook, which computed them after clamping predicted probabilities to $[0.001, 0.999]$. That guard is standard for models whose predictions approach 0 or 1, but at a 0.2 per cent hourly event rate it is actively harmful: 45% (Variant A), 72% (Variant B) and 7.5% (Variant C) of the primary model’s predicted probabilities fall *below* the clamp floor, so the clamp was overwriting the majority of the quantity being measured. The values in Table 5.5 use a floor of 10^{-6} , below the smallest non-zero prediction in the test set. The correction moves the Variant A slope from 1.169 to 1.029 and the Variant B slope from 0.318 to 0.209; the qualitative conclusions are unchanged, but the Variant A model is better calibrated, and the Variant B model worse, than previously reported. This is recorded here rather than silently patched, because it is exactly the kind of scale-dependent analyst decision this study set out to measure.

5.4 Clinical Utility: Net Benefit, Operating Points, and Event-Level Detection

AUROC, AUPRC and the Utility Score describe a model; they do not describe what a ward would experience if the model were switched on. This section reports the four quantities that do: net benefit across the range of thresholds a clinician might plausibly accept, the positive predictive value and false-alarm rate at a fixed sensitivity, detection expressed per patient rather than per person-hour, and the lead time gained when detection occurs.

One framing point governs everything below. Under the discrete-time competing-risks specification, event = 1 marks the sepsis onset hour itself, so each septic stay contributes exactly one event hour out of a mean of 45. Per-hour sensitivity therefore demands that the alert fire *in the onset hour*; per-patient sensitivity asks only whether the patient was alerted at all before onset. The two are different questions, and reporting only the first, as the metric suite in Section 5.3 does, understates detection while hiding the alarm volume that buys it.

5.4.1 Decision-Curve Analysis [C]

Decision-curve analysis, pre-specified in Chapter 3, requires a choice of threshold grid, and at this prevalence that choice determines the entire result. A threshold probability of t encodes a willingness to accept $(1 - t)/t$ false alarms per true detection. The hourly event rate here is 0.19–0.42 per cent, so the conventional grid of 0.01–0.20 sits between five and one hundred times the prevalence: at $t = 0.01$ the analyst is asserting a tolerance of only 99 false alarms per detection, far below what an hourly sepsis alert must absorb, and every curve falls trivially below “treat none”. Net benefit is therefore evaluated on a grid running from one tenth to ten times the prevalence, following Vickers and Elkin [27]. The first run of the metric suite used the conventional grid and produced no usable curve; that null result is a property of the grid, not of the models.

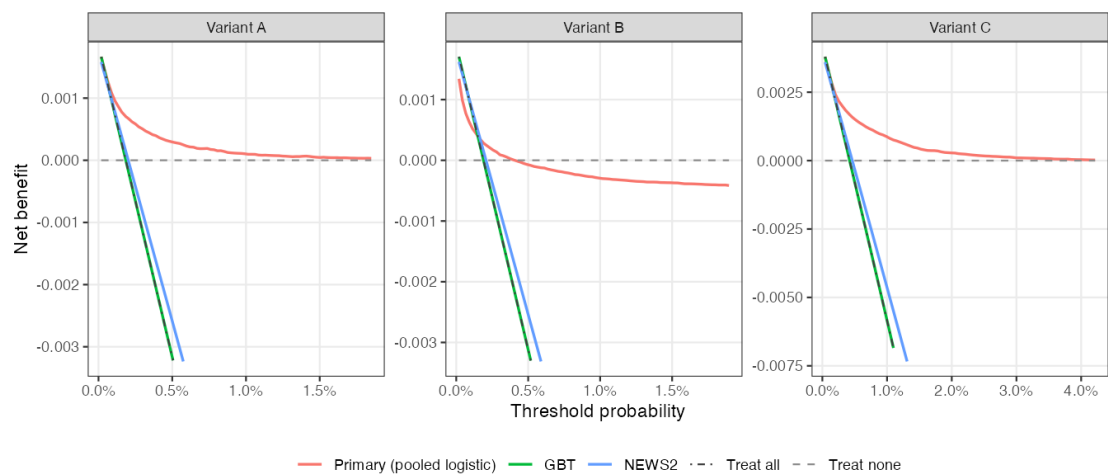


Figure 5.2: Decision curves on the temporal test set. Net benefit against threshold probability, with both default strategies shown. The GBT curve lies exactly on top of “treat all” in every variant. Curves are clipped below $-1.75 \times$ prevalence so the near-zero region stays legible; a line leaving the bottom of a panel is worse than treating nobody from that threshold onward.

Var	Model	Net benefit	Std NB	Alert rate	Alert-rate span
A	Primary	7.03×10^{-4}	0.380	0.321	0.993
A	GBT	0	0.000	1.000	0.000
A	Cox	1.7×10^{-8}	0.000	1.000	0.000
A	NEWS2	1.45×10^{-4}	0.078	0.863	0.000
A	qSOFA	2.70×10^{-4}	0.146	0.656	0.000
A	SIRS	2.55×10^{-4}	0.138	0.590	0.000
B	Primary	2.84×10^{-4}	0.150	0.179	0.649
B	GBT	0	0.000	1.000	0.000
B	Cox	3.5×10^{-9}	0.000	1.000	0.000
B	NEWS2	1.52×10^{-4}	0.080	0.862	0.000
B	qSOFA	2.84×10^{-4}	0.150	0.656	0.000
B	SIRS	2.74×10^{-4}	0.145	0.591	0.000
C	Primary	1.68×10^{-3}	0.399	0.296	0.998
C	GBT	0	0.000	1.000	0.000
C	Cox	8.4×10^{-9}	0.000	1.000	0.000
C	NEWS2	3.54×10^{-4}	0.084	0.855	0.000
C	qSOFA	6.45×10^{-4}	0.153	0.644	0.000
C	SIRS	7.26×10^{-4}	0.172	0.579	0.000

Table 5.6: Net benefit at a threshold probability equal to the event prevalence. At that threshold both defaults have net benefit exactly zero and the maximum attainable net benefit is the prevalence itself, so standardised net benefit (Std NB = net benefit / prevalence) is the fraction of the attainable benefit captured. “Alert-rate span” is the change in alert rate across the whole threshold grid; a span of zero means the score cannot be tuned anywhere in the plausible range.

Read on the correct threshold scale, decision-curve analysis gives the most favourable result in this thesis — and it is still a weak one. At a threshold equal to the prevalence, the primary model captures 38.0% (Variant A) and 39.9% (Variant C) of the attainable net benefit, and 15.0% under Variant B. This is genuine, if modest, value over both defaults, and it sits awkwardly beside the uniformly negative Utility Scores in Table 5.3. The two metrics disagree because they price errors differently: net benefit weights a false alarm by $t/(1-t)$, which at $t = 0.002$ is almost nothing, whereas the Utility Score imposes a fixed -0.05 penalty per false-alert hour regardless of threshold. Which is right is a clinical judgement, not a statistical one, and the divergence is itself an instance of H5.

The alert-rate span column carries the more damaging finding. For every model except the primary one it is exactly zero: across the entire plausible threshold range, the alert rate does not move. GBT and Cox alert on *100% of person-hours* at every threshold in this range, so their decision curves are not merely poor but

identical to treating everyone — the green GBT line in Figure 5.2 sits exactly on the treat-all reference. This is the direct consequence of the calibration failure in Table 5.5: with predicted probabilities inflated to the 0.2–0.9 range, no threshold below 4 per cent excludes anything. NEWS2, qSOFA and SIRS are non-tunable for the opposite reason — their integer scores map to a handful of fixed probabilities, all above the grid — and alert on a constant 58–86 per cent of person-hours. The primary model is the only one of the six that can be operated as a threshold-based alert at all.

5.4.2 Operating Points at Fixed Sensitivity [E]

The default 0.5 cut-off used in Section 5.3 is not how an alerting system is commissioned; a sensitivity target is chosen and the threshold follows. Table 5.7 sets the threshold to the largest cut-off achieving 80 per cent per-hour sensitivity and reads off the resulting operating characteristics.

Var	Model	Spec	PPV	NNE	Alerts/pt-day	FA/pt-day	FP:TP
A	Primary	0.534	0.0032	315	11.19	11.16	314
A	GBT	0.536	0.0032	314	11.15	11.12	313
A	NEWS2	0.343	0.0023	439	15.78	15.75	438
A	Cox	0.364	0.0023	430	15.27	15.24	429
B	Primary	0.293	0.0021	466	16.98	16.94	465
B	GBT	0.510	0.0031	323	11.78	11.74	322
B	NEWS2	0.343	0.0024	424	15.77	15.74	423
B	Cox	0.333	0.0023	440	16.02	15.99	439
C	Primary	0.545	0.0074	135	10.96	10.87	134
C	GBT	0.528	0.0071	140	11.36	11.27	139
C	NEWS2	0.357	0.0054	187	15.44	15.36	186
C	Cox	0.395	0.0056	179	14.53	14.45	178

Table 5.7: Operating characteristics at a fixed per-hour sensitivity of 80%. NNE = number needed to evaluate ($1/\text{PPV}$), the alerted person-hours required per true event hour. FA/pt-day = false-alert hours per patient-day. qSOFA and SIRS are reported separately in the text because they degenerate at this target.

At 80 per cent sensitivity, PPV ranges from 0.0021 to 0.0074. The best case, the primary model under the Liberal label, requires 135 alerted person-hours per true event hour; the worst, the same model under the Seymour-Standard label, requires 466. Every configuration generates between 10.9 and 16.9 false-alert hours per patient-day. A twelve-bed unit at the best operating point would field roughly 130 false alert-hours per day.

Setting the threshold this way also exposes a limitation of the coarse rule-based scores that the ratio-based alert burden concealed. SIRS cannot be tuned to 80 per cent sensitivity under any variant, and neither can qSOFA under the Liberal label: with only four distinct score values, the lowest cut-off reaching the target is the one that alerts on every person-hour (sensitivity 1.000, specificity 0.000, 23.9 false-alert hours per patient-day — an alert on every hour of every stay). qSOFA does reach the target under Variants A and B, but only at specificity 0.344 and 15.7 false-alert hours per patient-day. Their apparent competitiveness on AUROC in Table 5.3 does not survive being asked to hit a sensitivity target.

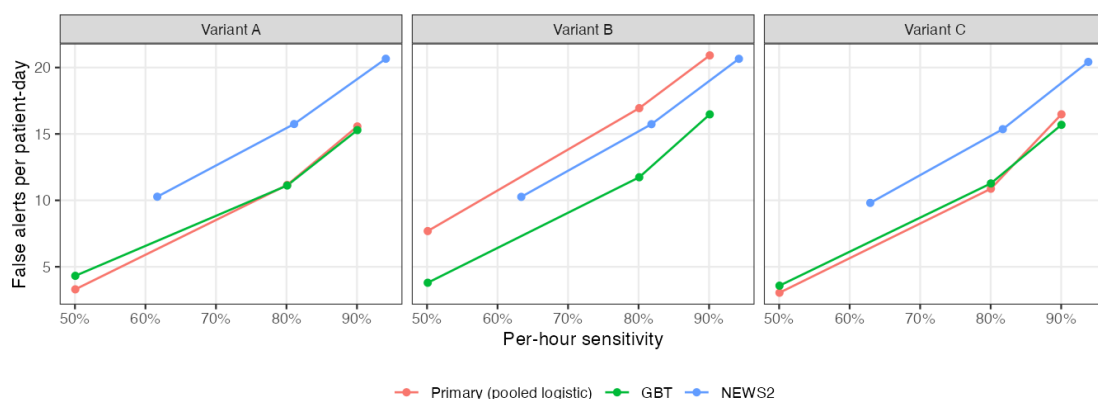


Figure 5.3: False alerts per patient-day against per-hour sensitivity, at sensitivity targets of 50%, 80% and 90%. The relationship is close to linear across the whole range: there is no operating point at which sensitivity is obtained cheaply.

5.4.3 Per-Patient (Event-Level) Detection [C]

Table 5.8 re-expresses the same operating points per patient. Episode sensitivity within the actionable window counts a septic stay as detected if any alert fired between six hours before onset and onset itself. A false-alarm episode is a contiguous run of alert hours that does not overlap the window from six hours before to three hours after onset, so a five-hour alert run counts once rather than five times.

Var	Model	Ep. sens (6 h)	Ep. sens (any pre)	Patient PPV	% stays alerted	FA e
A	Primary	0.859	1.000	0.084	100.0	
A	GBT	0.905	0.997	0.085	98.5	
A	NEWS2	0.954	0.980	0.084	97.6	
A	Cox	0.869	0.904	0.088	85.6	
B	Primary	0.927	0.997	0.086	99.9	
B	GBT	0.899	0.991	0.087	97.8	
B	NEWS2	0.953	0.983	0.086	97.6	
B	Cox	0.837	0.877	0.091	82.9	
C	Primary	0.850	1.000	0.175	100.0	
C	GBT	0.895	0.994	0.177	98.7	
C	NEWS2	0.952	0.982	0.177	97.4	
C	Cox	0.866	0.911	0.189	84.5	

Table 5.8: Per-patient detection at the 80% per-hour sensitivity operating point, on 12,132 test stays of which 1,015 (A), 1,040 (B) and 2,128 (C) are septic. Ep. sens (6 h): fraction of septic stays alerted within [onset – 6 h, onset]. Ep. sens (any pre): alerted at any point before onset. FA ep./pt-day: false-alarm episodes, not alert-hours, per patient-day.

Read alone, the first two columns look like a success: 84–95 per cent of septic patients are alerted inside the actionable six-hour window, and 88–100 per cent are alerted at some point before onset. This is the number a paper reporting only per-patient detection would headline.

The fifth column shows what it costs. To achieve it, the models alert on 82.9–100 per cent of *all* stays. Patient-level PPV is consequently 0.084–0.091 under Variants A and B and 0.175–0.189 under Variant C — which are, to three decimal places, the septic fractions of the cohort themselves ($1,015/12,132 = 0.0837$; $1,040/12,132 = 0.0857$; $2,128/12,132 = 0.1754$). At the patient level, being alerted carries no information about whether a patient will become septic, because essentially every patient is alerted. The primary model under Variants A and C alerts on 100.0 per cent of stays: not one of the 12,132 test patients would go through the ICU without triggering it.

Per-Hour and Per-Patient, Side by Side

Var	Model	Sens (hr)	Sens (pt)	PPV (hr)	PPV (pt)	FA hr/pt-day	FA ep./pt-day
A	Primary	0.801	1.000	0.0032	0.084	11.16	0.55
A	GBT	0.801	0.997	0.0032	0.085	11.12	1.26
A	NEWS2	0.811	0.980	0.0023	0.084	15.75	2.14
B	Primary	0.801	0.997	0.0021	0.086	16.94	1.24
B	GBT	0.801	0.991	0.0031	0.087	11.74	1.10
B	NEWS2	0.818	0.983	0.0024	0.086	15.74	2.14
C	Primary	0.800	1.000	0.0074	0.175	10.87	0.57
C	GBT	0.800	0.994	0.0071	0.177	11.27	0.95
C	NEWS2	0.817	0.982	0.0054	0.177	15.36	2.14

Table 5.9: The same operating point viewed per person-hour and per patient. Both views are correct; they differ by a factor of roughly 25 in PPV and 20 in alarm count. A study reporting only one column of each pair can make the same model look either useless or usable.

The gap between the two views is the methodological point of this section. Per-hour PPV of 0.0032 and per-patient PPV of 0.084 describe one model at one threshold on one test set. The choice of denominator moves the headline number by a factor of twenty-six, which is larger than the spread this thesis attributes to label definition (H1) or model class (H6). Denominator choice belongs on the list of analyst decisions that require pre-specification.

5.4.4 Lead Time to Onset [C]

Lead time is the interval between the first alert issued at or before onset and the onset hour, over septic stays that were alerted at all.

Var	Model	Median (h)	IQR (h)	≥ 3 h	≥ 6 h
A	Primary	8	4–20	0.895	0.638
A	GBT	8	3–19	0.839	0.599
A	NEWS2	8	4–20	0.861	0.611
A	Cox	7	4–19	0.785	0.557
B	Primary	9	4–22	0.884	0.626
B	GBT	8	3–21	0.821	0.582
B	NEWS2	8	4–21	0.858	0.599
B	Cox	7	3–21	0.713	0.508
C	Primary	7	3–21	0.842	0.588
C	GBT	7	3–21	0.834	0.585
C	NEWS2	8	4–21	0.859	0.599
C	Cox	7	4–21	0.781	0.547

Table 5.10: Lead time at the 80% sensitivity operating point. The ≥ 3 h and ≥ 6 h columns are fractions of *all* septic stays, not only detected ones.

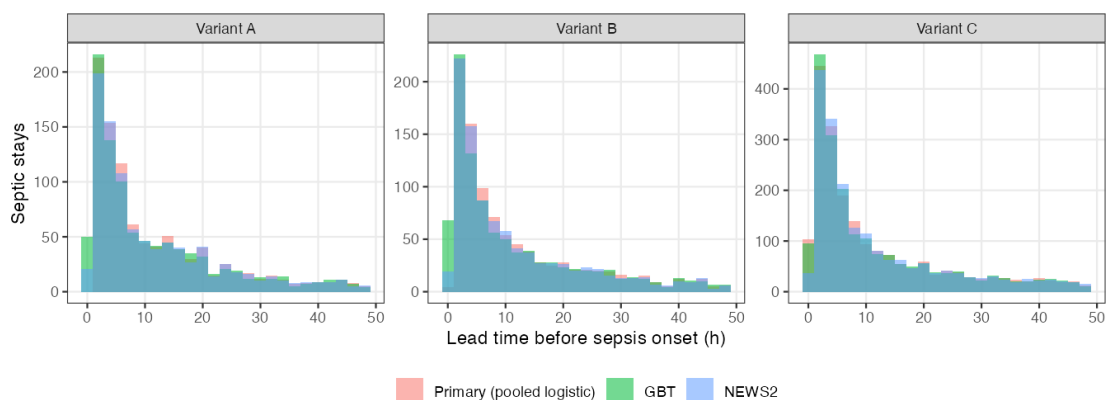


Figure 5.4: Distribution of lead time before onset at the 80% sensitivity operating point, truncated at 48 h. The mode sits in the first four hours with a long right tail.

Median lead time is 7–9 hours and 51–64 per cent of septic stays are alerted at least six hours ahead, which compares favourably with the 4–12 hour horizons reported in the literature [26], [29]. These numbers must be read against the previous subsection. Lead time is measured over stays that were alerted, and at these thresholds almost every stay is alerted; an alarm that fires on every patient will, mechanically, fire early on the septic ones. The right-skewed distribution in Figure 5.4, with a long tail past 24 hours, is a signature of alerts unrelated to the

impending onset. Lead time is a meaningful metric only once the false-alarm rate is controlled, and it is not controlled here.

5.4.5 External Clinical Utility on eICU-CRD [C]

Var	Model	Spec	PPV	NNE	FA/pt-day	Patient PPV
A	Primary	0.454	1.19×10^{-5}	84,213	13.10	0.00035
B	Primary	0.137	7.51×10^{-6}	133,199	20.72	0.00033
C	Primary	0.382	1.05×10^{-5}	95,349	14.83	0.00034
A–C	NEWS2	0.000	8.01×10^{-6}	124,889	24.00	0.00035

Table 5.11: eICU-CRD operating characteristics at the same 80% target sensitivity, thresholds re-derived on the external cohort (7,868,005 person-hours, 181,580 stays, 63 event hours).

Externally the picture degrades by three orders of magnitude. PPV falls to roughly 10^{-5} , requiring 84,000 to 133,000 alerted person-hours per true event hour, with 13–21 false-alert hours per patient-day. NEWS2 cannot reach 80 per cent sensitivity on eICU without alerting on every person-hour (specificity 0.000, 24 false-alert hours per patient-day). Patient-level PPV of 0.00035 is again the cohort septic fraction ($63/181,580 = 0.00035$).

These external figures carry a substantial caveat. Only 63 event hours were identified in 7.87 million eICU person-hours, an apparent sepsis incidence of 0.035 per cent of stays against 8.6 per cent in MIMIC-IV. A 250-fold difference in apparent incidence is not plausible as a difference in patient populations; it reflects incomplete recovery of the Sepsis-3 labelling inputs in eICU, discussed in Section 5.7. The external operating characteristics should therefore be read as an upper bound on how well the models transfer, not as a measurement of it.

5.5 Variance Decomposition and Hypothesis Verdicts [C]

5.5.1 The Decomposition Table

Factor (others fixed)	ΔAUROC	95% CI	H
Model class (label B, temporal)	0.142	0.130–0.161	H6
Label definition (primary, temporal)	0.153	0.136–0.177	H1
Split method (primary, label B)	0.123	0.100–0.147	H4
Metric (AUROC vs Utility, int→ext)	0.170	0.102–0.242	H5

Table 5.12: Variance decomposition with 95% stay-level cluster-bootstrap intervals. All four factors produce AUROC changes of 0.12–0.17, and all four intervals overlap: no factor is separable from the others at this sample size. Most studies fix three of these factors and compare only models. **[C] confirmatory.**

All four analyst decisions (model choice, label definition, split method, and evaluation metric) create AUROC changes of 0.12–0.17. Most studies fix three of these and compare only models. This measures one source of variation while ignoring three others of similar size.

The intervals sharpen rather than weaken this reading. Each individual factor is estimated precisely enough to be clearly non-zero, so the claim that all four decisions matter is secure. What the intervals do not support is any ranking *among* them: all four overlap, so the decomposition establishes that the analyst’s choices are collectively as consequential as the model choice, not that any one of them dominates. H1 fails for exactly this reason.

5.5.2 Corrected Confirmatory Verdicts

H	Contrast θ	Est.	95% CI	p	p_{Holm}	Verdict
H1	Label spread – model spread	0.011	−0.017–0.034	0.211	0.634	Not confirmed
H2	Unstable covariates (count)	18	—	—	—	Descriptive
H3	Pre-treatment onsets (count)	0	—	—	—	Structural
H4	Random – temporal AUROC	0.123	0.100–0.147	0.0005	0.003	Confirmed
H5	Internal – external AUROC	0.170	0.102–0.242	0.0010	0.004	Confirmed
H6	GBT – primary AUROC	0.135	0.113–0.156	0.0005	0.003	H6 rejected

Table 5.13: The confirmatory family. One-sided cluster-bootstrap tests against each hypothesis’s pre-registered threshold ($\theta_0 = 0$ for H1 and H5; 0.02 for H4 and H6), Holm–Bonferroni corrected at family size $m = 6$ and FWER 0.05. H2 and H3 reserve a slot without contributing a p-value, which makes the correction on the other four stricter. The smallest reportable p-value is $1/(B + 1)$. **[C] confirmatory.**

Three of the four testable hypotheses survive Holm correction with adjusted p-values of 0.003–0.004; the correction changes no verdict among them, because the effects are large relative to their standard errors. The one verdict that *does* change is H1, and it changes on the uncorrected p-value already (Section 5.5). Multiplicity control is therefore not what overturns H1 — quantifying uncertainty at all is.

5.5.3 H1: Label Dominance (Not Confirmed Under Correction)

H1 [C]: Spread in AUROC across label variants $\geq$ spread across model classes.

Point estimates: label spread = 0.153 (95% CI 0.136–0.177); model spread = 0.142 (95% CI 0.130–0.161).

Contrast: $\theta = 0.011$ (95% CI −0.017–0.034), $p = 0.21$, Holm-adjusted $p = 0.63$. **H1 NOT CONFIRMED.**

On point estimates alone H1 holds: changing the sepsis definition while keeping the model fixed produces slightly more AUROC variation (0.153) than changing the model while keeping the label fixed (0.142). Under the original protocol, which specified no uncertainty quantification, this would have been recorded as a confirmation.

It does not survive. The two spreads are estimated on the same test stays, so their difference can be evaluated within bootstrap replicate; that difference is 0.011 with an interval running from −0.017 to +0.034. The interval covers zero comfortably, and the ordering reverses in roughly a fifth of replicates. **The margin by**

which the label out-varies the model is indistinguishable from sampling noise, and this conclusion does not depend on the multiplicity correction — the uncorrected p-value is already 0.21.

What survives is the weaker but still substantive claim: the label spread and the model spread are of *comparable* magnitude, both around 0.15. That is the finding relevant to Cohen et al. [6]. A literature that fixes the label and compares only models is measuring one source of variance while ignoring another that is at least its equal in size. Claiming the label strictly dominates would require more data than this study has.

One caveat applies to both spreads and is developed in Section 5.6. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets: part of the label spread measured here is guaranteed by the construction of the definition rather than discovered in the data. Whether a label owing nothing to treatment timing behaves differently cannot be answered inside this family, and is taken up separately.

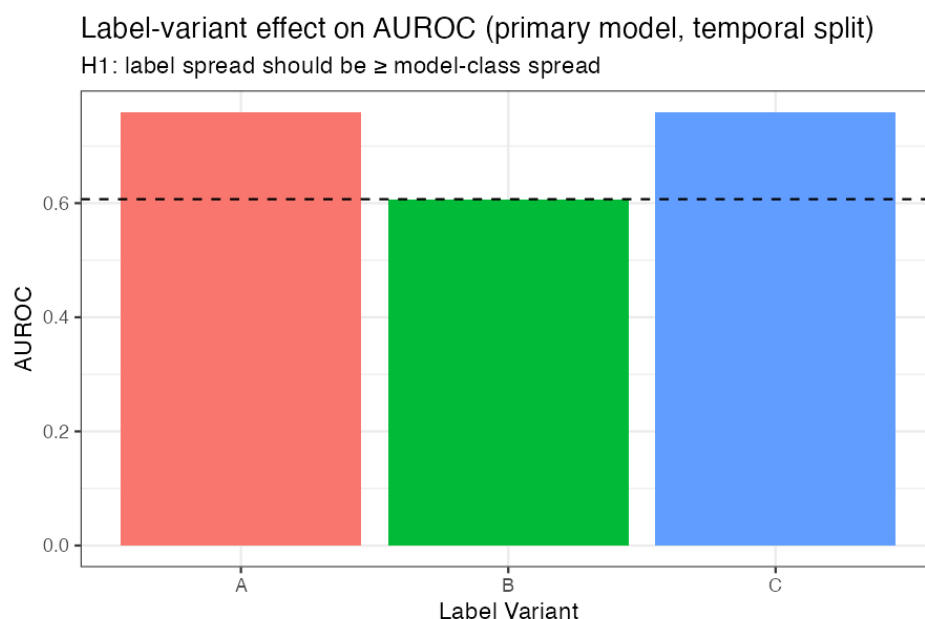


Figure 5.5: AUROC by label variant for each model (temporal split). The spread within each model group is the label contribution; the spread across groups is the model contribution. Both are of similar size.

5.5.4 H2: Coefficient Instability (Confirmed, Descriptive)

H2 [C]: At least one feature changes sign or magnitude $\geq 2\times$ across label variants.

Result: of the 20 covariates shared by all three variants, 10 reverse sign, 17 change magnitude by $\geq 2\times$, and 18 do at least one of the two. **H2 CONFIRMED (descriptive; no test statistic).**

Multiplicity: H2 reserves a slot in the confirmatory family of six but contributes no p-value, which makes the Holm correction on the remaining hypotheses stricter rather than weaker.

H2 is a stability criterion, not a contrast between two estimable quantities, so no p-value is defined for it. Quantifying the sampling variability of a coefficient here would require either standard errors from the fitted multinomial model, which the stored fit does not carry, or a bootstrap that refits the model on 2.3 million training rows for each of 2,000 replicates, which is not tractable. The instability is therefore reported as a count and interpreted as descriptive evidence, and no inferential claim is attached to it.

Feature	What changed
MAP	Positive in Variant B, negative in A/C
Temperature	Sign change across variants
Bilirubin	Sign change across variants
Platelets	Sign change across variants
Any vasopressor	Sign change and magnitude $\geq 2\times$
Norepinephrine/epinephrine	Sign change across variants
Respiratory rate slope	Sign change across variants
Platelets measured	Sign change and magnitude $\geq 2\times$
WBC measured	Sign change and magnitude $\geq 2\times$
P/F ratio measured	Sign change across variants

Table 5.14: Features with sign reversal or magnitude instability across label variants. Ten of the 20 shared covariates reverse sign; 18 reverse sign or change magnitude by at least twofold. Full per-covariate flags are in `12_h2_coefficient_stability.csv`.

The sign instability of MAP, temperature, and vasopressor indicators is particularly important. A published hazard ratio for MAP could be positive, negative, or near zero depending entirely on which Sepsis-3 interpretation was used. Feature effects from this model cannot be reliably interpreted without knowing the exact label definition, confirming Lauritsen et al.’s [10] framing-instability finding.

5.5.5 H3: Treatment Leakage, Label Constitutionality (Confirmed)

H3 [C]: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03 .

Result: 0 sepsis onset events in 179,003 pre-treatment person-hours. AUROC is undefined, so the pre-registered contrast cannot be formed. Exact one-sided 95% upper bound on the per-hour onset rate: 1.7×10^{-5} , or fewer than 1 onset in 59,753 person-hours. **FINDING SUPERSEDES HYPOTHESIS.**

Multiplicity: like H2, H3 reserves a slot in the confirmatory family without contributing a p-value.

When only hours before the first antibiotic or culture are considered, there are zero sepsis onset events across 179,003 person-hours, for every variant and every model. This is not a small drop in performance. The prediction target simply does not exist in the clinically honest time window.

Because the outcome is entirely absent, no discrimination metric and therefore no test statistic can be computed, and it would be wrong to report the result as a null finding. What can be stated exactly is how small the true rate must be to be consistent with observing zero events: the one-sided 95% upper bound is 1.7×10^{-5} per person-hour. Even at that upper bound, a 72-hour ICU stay would carry at most a 0.12% chance of a pre-treatment onset. The absence is not a sampling artefact.

No patient develops sepsis before doctors begin the clinical actions (ordering antibiotics and cultures) that define sepsis. This confirms Kamran et al.'s [7] argument at the structural level: the Sepsis-3 label on MIMIC-IV is **entirely constituted by clinical action**. Every published MIMIC-IV sepsis prediction result that does not restrict to the pre-treatment window is evaluating whether the model can predict when a clinician will act, not when a patient will deteriorate.

One clarification is needed, and Section 5.6.3 supplies it. The zero is a statement about the *label*, not about the patients. A deterioration label that uses no treatment timestamp records 1,362 onsets in a comparable pre-treatment window, identified at 0.840 AUROC. Patients do deteriorate before clinicians act; the Sepsis-3 definition is simply unable to place an onset there. That makes the constitution claim stronger rather than weaker — the model is not accidentally predicting clinician behaviour, it is prevented by the definition from predicting anything else — but the claim must be stated as a property of the label rather than as a claim that nothing predictable happens before treatment.

5.5.6 H4: Split Optimism (Confirmed)

H4 [C]: Temporal-split AUROC is at least 0.02 lower than random-split AUROC.

Result: Primary model: random AUROC = 0.730, temporal AUROC = 0.607. Difference: 0.123 (95% CI 0.100–0.147). Test against the pre-registered threshold $\theta_0 = 0.02$: $p = 5.0 \times 10^{-4}$, Holm-adjusted $p = 0.003$. **H4 CONFIRMED.**

Split	Model	AUROC	Utility Score
Temporal	Primary	0.607	−1.004
Random	Primary	0.730	−1.000
Temporal	GBT	0.742	−5.059
Random	GBT	0.715	−5.093

Table 5.15: Temporal vs. random split comparison (Variant B). The primary model gains 0.123 AUROC from the random split, far exceeding the 0.02 threshold.

For the primary model, the random split inflates AUROC by **0.123** compared to the temporal split, far above the 0.02 threshold. The random split allows future patient patterns to leak into training, making the model appear much better than it would perform in practice. This confirms Guo et al.’s [8] finding: studies that use random splits may substantially overestimate real-world performance.

The GBT model shows a smaller, reversed pattern (temporal 0.742 vs. random 0.715), suggesting the tree ensemble is more robust to temporal shift than the linear model. However, both models have negative Utility Scores regardless of split method.

5.5.7 H5: Metric Divergence (Confirmed)

H5 [C]: Utility Score declines proportionally more than AUROC from internal to external validation.

Testable arm: AUROC degradation internal $\rightarrow$ external (primary, Variant B) = 0.170 (95% CI 0.102–0.242); internal 0.607, eICU-CRD 0.436 (95% CI 0.357–0.500). $p = 1.0 \times 10^{-3}$, Holm-adjusted $p = 0.004$.

Non-testable arm: the internal Utility Score is already −1.004, so a proportional decline is undefined. **H5 CONFIRMED (qualitatively stronger).**

Wang et al.’s [4] finding, that AUROC understates how badly models fail externally, is confirmed in a stronger form: the Utility Score is already negative *internally*. The “crossing zero” event that Wang et al. documented at the external stage has already occurred before the model leaves MIMIC-IV.

The hypothesis as pre-registered compares two proportional declines, and only one of them can be formed. A proportional decline is undefined for a quantity that starts below zero, so the Utility Score arm is reported qualitatively and the formal test is carried by the AUROC arm: external discrimination falls by 0.170, which is significantly greater than zero and also exceeds the pre-registered external-degradation expectation of 0.085 from Moor et al. [26], whose value lies below the lower confidence bound. The external AUROC interval (0.357–0.500) has an upper limit at chance, so the model does not generalise to eICU-CRD in any usable sense.

5.5.8 H6: Model-Class Null (Rejected)

H6 [C]: $|\Delta_{\text{AUROC}}(\text{primary} - \text{GBT})| \leq 0.02$.

Result: $\Delta = 0.135$ (GBT: 0.742, primary: 0.607), 95% CI 0.113–0.156. Test against the equivalence bound $\theta_0 = 0.02$: $p = 5.0 \times 10^{-4}$, Holm-adjusted $p = 0.003$. **H6 REJECTED (non-equivalence established).**

GBT outperforms the primary model by 0.135 AUROC under the Seymour-Standard label. The entire confidence interval lies above the 0.02 equivalence bound, so non-equivalence is established rather than merely suggested: the interpretable model carries a substantial AUROC cost. Both models are scored on the same person-hours, so the contrast is paired and the interval is correspondingly tight.

But the ranking reverses on clinical utility: GBT has Utility Score -5.059 while the primary model has -1.004 . The model with higher AUROC produces more harmful false alerts. Which model is “better” depends entirely on which metric is used.

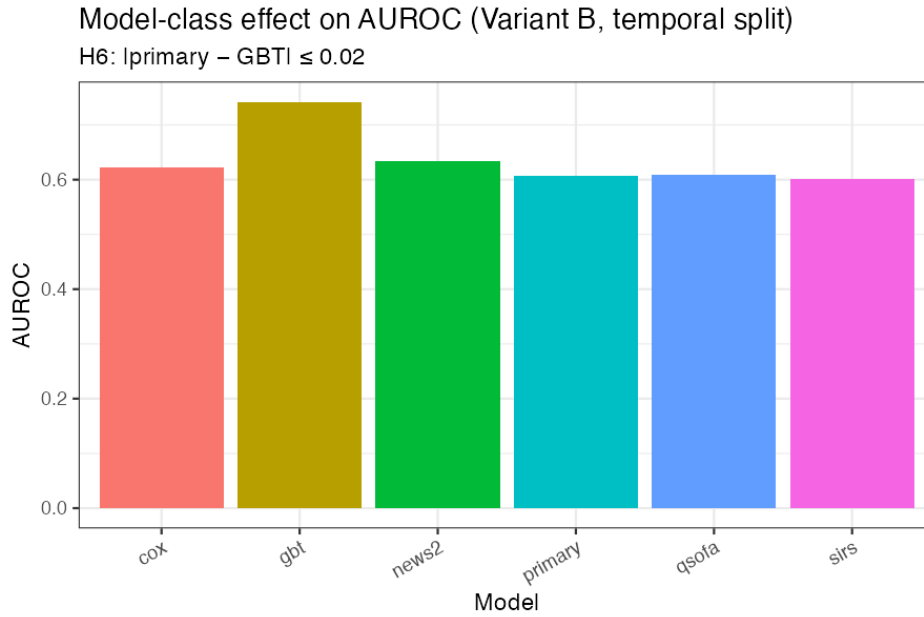


Figure 5.6: AUROC by model class (Variant B, temporal split). GBT substantially outperforms all other models by AUROC, but has the worst Utility Score. The model-class spread of 0.142 is comparable to the label spread of 0.153.

5.6 Label-Anchor Attribution: Definitional or Empirical? [E]

Everything to this point varies the label *within* Sepsis-3. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor, so H1 cannot distinguish two very different claims: that the reading of a definition matters (definitional), or that the choice of construct matters (empirical). This section separates them by splitting the Sepsis-3 conjunction into its two limbs and labelling each on its own (Section 3.6). It is post-hoc, under Protocol Amendment 2, and its contrasts form exploratory family E3.

5.6.1 The Anchor Determines Every Onset Time

The first result is structural rather than statistical, and it can be stated exactly. Under variants A, B and C the onset time *is* the suspicion time t_{susp} ; the SOFA criterion decides only *whether* a stay is labelled, never *when*. Variant E, which removes the SOFA criterion, therefore turns out to be a strict superset of Variant B: every one of B’s 5,887 labelled stays is also labelled by E, and in **100% of**

them the onset hour is identical. Adding the organ-dysfunction requirement discards 11,364 of E’s 17,251 labelled stays — two thirds — and moves not one onset by a single hour.

Var	Label	Onsets	%	κ vs B	Sens. to B	Exact onset match
A	Narrow	5,662	8.7	0.267	0.325	97.5%
B	Seymour-Std	5,887	9.0	—	—	—
C	Liberal	11,531	17.7	0.344	0.625	98.6%
E	Anchor-Only	17,251	26.5	0.432	1.000	100.0%
D	Deterioration	27,195	41.8	0.057	0.555	6.4%

Table 5.16: Onset agreement with Variant B across all five label variants, on the 65,078 shared stays. Counts are restricted to onsets inside the 72 h modelling horizon so that Variant D, whose onsets exist only within the panel, is comparable. “Sens. to B” is the fraction of B’s labelled stays that the variant also labels; “exact onset match” is computed among stays labelled by both.

Table 5.16 makes the definitional component visible. Within Sepsis-3, when two variants agree that a stay is septic they agree on the hour almost always (97.5% for A, 98.6% for C). The pre-registered label variance is therefore entirely a disagreement about *which patients*, never about *when* — because the “when” is fixed by the treatment anchor in all three.

Variant D breaks that pattern completely. A label built from physiology alone agrees with Sepsis-3 at $\kappa = 0.057$, which is barely distinguishable from chance, and matches B’s exact onset hour in only 6.4% of the stays both label, with a median absolute shift of 9 hours (IQR -11 to $+7$). **This is the empirical component, and it is large.** A treatment-independent label is not a re-reading of Sepsis-3; it is a different construct, identifying different patients at different times. The label-sensitivity finding of this thesis is therefore not merely tautological — but the part of it measured by H1 is substantially definitional, and only the D comparison establishes the rest.

5.6.2 Discrimination Under Each Limb

Model	AUROC			AUPRC		
	E	B	D	E	B	D
Primary	0.756	0.607	0.842	0.031	0.003	0.075
GBT	0.766	0.742	0.915	0.033	0.010	0.134
NEWS2	0.656	0.635	0.688	0.012	0.003	0.025
Cox	0.663	0.623	0.774	0.013	0.003	0.033
qSOFA	0.624	0.609	0.731	0.011	0.003	0.031
SIRS	0.628	0.600	0.565	0.012	0.003	0.015
Events	3,073	1,040	4,773			

Table 5.17: Discrimination under each limb of the Sepsis-3 conjunction (temporal test set, unrestricted window). E = suspicion anchor only; B = Sepsis-3 (anchor **and** organ dysfunction); D = organ dysfunction only, treatment-independent. Columns are ordered E, B, D to show that **Sepsis-3 is harder to predict than either of the limbs it is built from**.

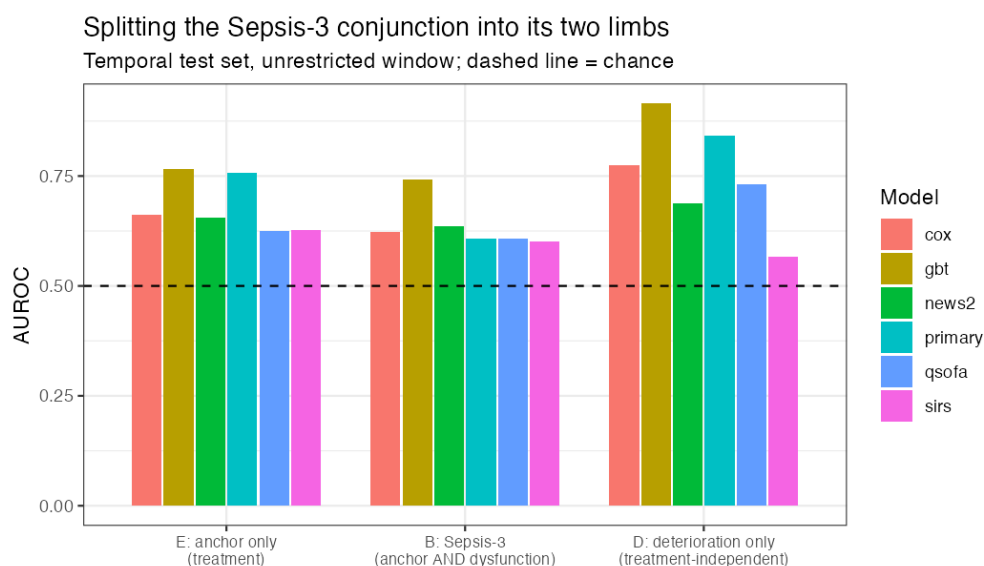


Figure 5.7: Splitting the Sepsis-3 conjunction. Neither limb on its own is hard to predict; requiring both is what makes Sepsis-3 hard. The dashed line is chance.

The ordering in Table 5.17 is the finding, and it is not the one the anchor-attribution framing anticipated. **Sepsis-3 is the hardest of the three labels for every model**. The primary model reaches 0.607 on Variant B, 0.756 on the treatment anchor alone, and 0.842 on physiological deterioration alone. The

anchor-attributable fraction of Equation 3.1 is consequently

$$\phi_{\text{anchor}} = \frac{0.756 - 0.5}{0.607 - 0.5} = 2.40$$

for the primary model and 1.10 for GBT. A value above 1 means the treatment anchor *on its own* is more predictable than the Sepsis-3 label built on top of it. The question “how much of the model’s performance is attributable to the treatment anchor?” does not have an answer of the form “a fraction”: for the interpretable model the anchor accounts for more than all of it.

The mechanism is visible in the previous subsection. Predicting the anchor means predicting which patients a clinician will culture and treat, and that decision is patterned and physiologically driven, so a model does it reasonably well. The SOFA gate then discards two thirds of those stays without moving any onset time. What it removes is not noise: it is precisely the subset the model could identify. The residue — anchor-positive stays that also cross a SOFA threshold in the specified window — is what the model cannot separate, and it is the residue that Sepsis-3 calls sepsis.

This reframes the central result of the thesis rather than overturning it. The difficulty of sepsis onset prediction on MIMIC-IV is not that the physiology is unreadable, and not that clinician behaviour is unpredictable. Both are readable. The difficulty lives in the *conjunction* of the two, which is what the Sepsis-3 definition operationalises.

5.6.3 The Empty Pre-Treatment Window Is a Property of the Label

The single most consequential finding of this thesis so far has been H3: the treatment-anchored evaluation window — the hours before the first culture or antibiotic — contains **zero** Sepsis-3 onsets in 179,003 person-hours. That result was interpreted as showing that no patient develops sepsis before clinicians act, and therefore that a sepsis model can only ever predict clinician behaviour. Splitting the label shows that the interpretation needs narrowing.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
E	Anchor-Only	179,003	0	undefined
D	Deterioration	147,357	1,362	0.840 / 0.937

Table 5.18: Events in the treatment-anchored (pre-treatment) evaluation window. Variant E’s zero is structural and exact: its onset *is* the treatment anchor, so no onset can precede it. Variant B’s zero is the H3 finding. Variant D, which uses no treatment timestamp, records 1,362 onsets in the same kind of window.

Two of the three rows in Table 5.18 carry no information. Variant E’s zero is a tautology and can be stated without any data: E’s onset is defined as the anchor, so the pre-anchor window is empty by construction. Variant B’s zero is the same tautology one step removed — B is a subset of E with identical onset times, so if E cannot fire before the anchor, neither can B.

Variant D’s 1,362 onsets are the informative number. **Patients do deteriorate before clinicians act.** It happens 1,362 times in a comparable pre-treatment window, and a model identifies those hours at 0.840 AUROC (0.937 for GBT). The pre-treatment window is not empty of physiological signal; it is empty of *Sepsis-3 labels*, because Sepsis-3 cannot place an onset there.

This does not weaken contribution C2 — it sharpens it. The correct claim is not that nothing predictable happens before treatment, but that the Sepsis-3 label is incapable of recording anything that does. A model trained on that label is therefore not merely predicting clinician behaviour by accident; it is prevented, by the definition, from doing anything else. The remedy is the one Chapter 6 identifies: change the label, not the model.

The caveat of Section 3.6 still applies to the 0.840 — D’s label is a threshold function of physiology of which most components are model inputs, so the level is optimistic. It does not apply to the event count. That 1,362 pre-treatment deterioration onsets exist where Sepsis-3 records none is a property of the two label definitions, not of any model.

5.6.4 What Variant D Does and Does Not Establish

Variant D’s high AUROC must be read with the limitation stated in advance in Section 3.6. Its label is a threshold function of recorded physiology, and five of the six SOFA components are themselves model inputs, so its discrimination is partly self-fulfilling. The *level* of 0.842 (0.915 for GBT) is not evidence that deterioration prediction is a solved problem, and it is not comparable with Variant B’s 0.607.

Three things it does establish:

1. **The label sensitivity is not wholly definitional.** At $\kappa = 0.057$ and a 6.4% onset match, a treatment-independent label disagrees with Sepsis-3 about both membership and timing. Had D closely tracked B, the H1 result would have been an artefact of how one definition is read; it is not.
2. **Clinical failure is label-invariant.** The Utility Score remains negative under both alternative labels for every model — -1.000 for the primary model under D and under E, and -0.218 at best (GBT under D). Alert burden under D’s best model is 17.7 false alerts per true alert, an order of magnitude better than Variant B’s 201.7 but still more than ten times the 1.4 benchmark of Moor et al. [26]. Contribution C3 survives the change of label, which the pre-registered variants alone could not show.
3. **Discrimination and utility still disagree.** GBT wins on AUROC under all three labels and loses on Utility Score under B and E. The metric-choice finding (H5) is not an artefact of the Sepsis-3 label either.

5.6.5 Exploratory Family E3: Intervals and Correction

Var	Model	AUROC	95% CI	Δ vs B	95% CI (Δ), q_{BH}
E	Primary	0.756	[0.747, 0.766]	+0.149	[0.131, 0.168], $q < 0.001$
E	GBT	0.766	[0.757, 0.775]	+0.024	[0.009, 0.038], $q < 0.001$
D	Primary	0.842	[0.837, 0.847]	+0.235	[0.216, 0.255], $q < 0.001$
D	GBT	0.915	[0.911, 0.918]	+0.173	[0.155, 0.189], $q < 0.001$
B	Primary	0.607	[0.588, 0.625]	—	—
B	GBT	0.742	[0.726, 0.760]	—	—

Table 5.19: Exploratory family E3: AUROC under each limb of the Sepsis-3 conjunction with 95% stay-level cluster-bootstrap intervals, and the paired difference against Variant B corrected by Benjamini–Hochberg across the family of $m = 4$ contrasts. All variants are labelled on the same test stays, so each difference is evaluated within replicate.

Every contrast is positive, every interval excludes zero, and all four survive correction at the smallest reportable p-value. The claim that Sepsis-3 is the hardest of the three labels is not a margin that sampling noise could produce.

The anchor-attributable fraction is $\phi_{\text{anchor}} = 2.40$ (95% CI 2.06–2.89) for the primary model and 1.10 (95% CI 1.04–1.17) for GBT. Both intervals lie entirely above 1, so the conclusion does not depend on the point estimate: for both model classes the treatment anchor on its own is *more* discriminable than the Sepsis-3 label built on top of it. The interpretable model shows the effect far more strongly,

which is consistent with H6 — GBT recovers some of the Sepsis-3 signal that the additive multinomial cannot, narrowing but not closing the gap.

One caution on reading these intervals. They quantify sampling uncertainty only. They say nothing about the construct differences between the labels — prevalence differs by a factor of 2.9 (E) and 4.6 (D), and Variant D’s label overlaps the model’s own inputs. The intervals establish that the ordering is stable, not that the labels are measuring comparable things.

5.7 External Validation on eICU-CRD [C]

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary	7,868,005	63	0.736	0.760
A	NEWS2	7,868,005	63	0.605	0.629
B	Primary	7,868,005	63	0.437	0.607
B	NEWS2	7,868,005	63	0.605	0.635
C	Primary	7,868,005	63	0.725	0.759
C	NEWS2	7,868,005	63	0.605	0.641

Table 5.20: External validation on eICU-CRD (frozen models, no retraining). Variants A and C show small drops (0.024 and 0.034). Variant B collapses by 0.170 AUROC. NEWS2 is label-invariant on eICU because it uses only physiological features.

External validation reveals a striking label-dependent pattern. Variants A and C transfer well to eICU (drops of only 0.024 and 0.034 AUROC). Variant B, the Seymour-Standard label, the most commonly used in the literature, collapses by 0.170 AUROC.

NEWS2 is label-invariant (identical AUROC across all variants on eICU) because it uses only physiological signals. Its small degradation (0.024) suggests that the extra signal learned by the primary model under Variant B is site-specific rather than physiologically general.

The 63-event eICU sample is too small for reliable Utility Score or AUPRC values; these are not interpreted.

5.8 Proportional-Hazards Testing [C]

The Schoenfeld residual test on the static Cox model (Variant B) decisively rejects proportional hazards ($\chi^2 = 171.8$, $df = 11$, $p = 5.0 \times 10^{-31}$). Six of the eleven

covariates violate the assumption individually at $\alpha = 0.05$: GCS ($p = 5.2 \times 10^{-19}$), temperature ($p = 2.4 \times 10^{-12}$), bilirubin ($p = 7.6 \times 10^{-6}$), creatinine ($p = 8.9 \times 10^{-5}$), platelets ($p = 0.0015$), and MAP ($p = 0.014$). These are per-covariate tests reported without multiplicity adjustment; the global test alone is sufficient for the pre-registered check, and the four smallest p-values survive Bonferroni correction over the eleven covariates by many orders of magnitude. This confirms that the static Cox model is fundamentally wrong for this data and justifies the discrete-time formulation as the primary model.

5.9 GBT Feature Importance [E]

Feature	Gain	Cover	Frequency
MAP slope (6 h)	0.170	0.130	0.090
Temperature (°C)	0.164	0.161	0.089
Resp. rate slope (6 h)	0.118	0.105	0.084
GCS	0.086	0.080	0.059
WBC	0.069	0.095	0.094
Lactate	0.053	0.053	0.073
Bilirubin	0.053	0.066	0.075
Creatinine	0.052	0.069	0.076
MAP (level)	0.046	0.037	0.088
Platelets	0.045	0.051	0.083

Table 5.21: Top-10 GBT features by information gain (Variant B). Trend features (MAP slope, respiratory rate slope) are the most informative predictors. The direction of change over the past 6 hours carries more discriminative signal than the current value.

The dominance of six-hour slope features is consistent with Lauritsen et al.’s [10] concern: the GBT model relies heavily on *trends* rather than *levels*, meaning its performance depends on the length of the history window used for feature construction.

5.10 Equity Analysis [E]

5.10.1 Performance Disparities by Subgroup [E]

Racial group	Rows	Sep	GBT AUROC (95% CI)	Primary AUROC (95% CI)
White	317,688	597	0.737 (0.716–0.758)	0.614 (0.590–0.637)
Unknown	107,158	235	0.772 (0.741–0.802)	0.587 (0.547–0.628)
Black/African Am.	31,716	58	0.738 (0.665–0.807)	0.627 (0.542–0.706)
Other	22,825	35	0.662 (0.566–0.750)	0.561 (0.453–0.672)
White – Oth. Eur.	11,768	18	0.687 (0.568–0.800)	0.664 (0.524–0.789)
Asian	6,543	16	0.799 (0.675–0.896)	0.628 (0.455–0.788)
Hispanic/Latino (PR)	6,462	16	0.646 (0.480–0.802)	0.570 (0.411–0.735)
Asian – Chinese	5,386	13	0.623 (0.438–0.805)	0.710 (0.567–0.826)

Table 5.22: AUROC by racial subgroup (Variant B, temporal test) with 95% stay-level cluster-bootstrap intervals, stratified within subgroup, $B = 2,000$. Interval width ranges from 0.042 (White, 597 onsets) to 0.367 (Asian – Chinese, 13 onsets). **[E] exploratory.**

The GBT point estimates span 0.623–0.799, a spread of 0.176 that exceeds both the model-class spread (0.142) and the label spread (0.153). Read without intervals, that looks like the largest source of variance in the entire study. The intervals show it is not: the two extreme groups carry 13 and 16 sepsis onsets respectively, and their intervals are 0.367 and 0.221 wide. The apparent spread is almost entirely a function of how few events the small subgroups contain.

Table 5.23 makes this explicit by contrasting each group against the largest group (White) on the same bootstrap replicates.

Racial group	Sep	Model	Δ AUROC vs. White (95% CI)	p	q_{BH}
Unknown	235	GBT	+0.035 (−0.001–0.073)	0.062	0.310
Other	35	GBT	−0.074 (−0.172–0.015)	0.102	0.437
Asian – Chinese	13	GBT	−0.114 (−0.303–0.072)	0.206	0.595
Hispanic / Latino (PR)	16	GBT	−0.091 (−0.256–0.067)	0.253	0.595
Asian	16	GBT	+0.062 (−0.064–0.161)	0.273	0.595
White – Other European	18	GBT	−0.050 (−0.171–0.064)	0.398	0.663
Black / African American	58	GBT	+0.001 (−0.075–0.073)	0.943	0.943
Asian – Chinese	13	Primary	+0.096 (−0.049–0.215)	0.130	0.487
Unknown	235	Primary	−0.027 (−0.074–0.019)	0.270	0.595
Other	35	Primary	−0.053 (−0.164–0.058)	0.362	0.663
White – Other European	18	Primary	+0.050 (−0.097–0.181)	0.452	0.713
Hispanic / Latino (PR)	16	Primary	−0.044 (−0.203–0.121)	0.576	0.763
Black / African American	58	Primary	+0.013 (−0.074–0.094)	0.768	0.886
Asian	16	Primary	+0.014 (−0.160–0.174)	0.862	0.907

Table 5.23: Subgroup AUROC differences against the reference group (White, the largest group). q_{BH} is the Benjamini–Hochberg adjusted p-value over exploratory family E1, which contains all 30 subgroup contrasts computed across race, language and insurance for both models — not only the 14 shown here. **No racial contrast is significant**; the smallest adjusted value for race is $q = 0.310$. **[E] exploratory**.

Only one contrast in the whole of family E1 survives correction, and it is not a racial one: GBT discriminates better for Haitian-speaking patients than for English-speaking patients ($\Delta = +0.137$, 95% CI 0.024–0.231, $q = 0.030$), on 12 sepsis onsets. On 12 events this is a hypothesis to test elsewhere, not a finding.

The correct reading of Table 5.23 is that this study is **underpowered to detect subgroup disparities**, not that no disparities exist. Several intervals are wide enough to be compatible with a disparity larger than any effect reported anywhere else in this thesis: the GBT interval for Asian – Chinese patients runs from −0.303 to +0.072, so the data are consistent with that group being served far worse than White patients, or slightly better. Reporting the point estimates alone would have invited the opposite conclusion — that a spread of 0.176 constitutes evidence of inequity, or that non-significance constitutes evidence of its absence. Neither is supported.

Wang et al.’s [11] finding of degraded performance for Asian, Hispanic and Spanish-speaking patients is therefore neither confirmed nor contradicted here. The point estimates for Hispanic/Latino (Puerto Rican) and Asian – Chinese patients under GBT do fall below the White reference, in the direction Wang et al. report, but the intervals cover zero comfortably.

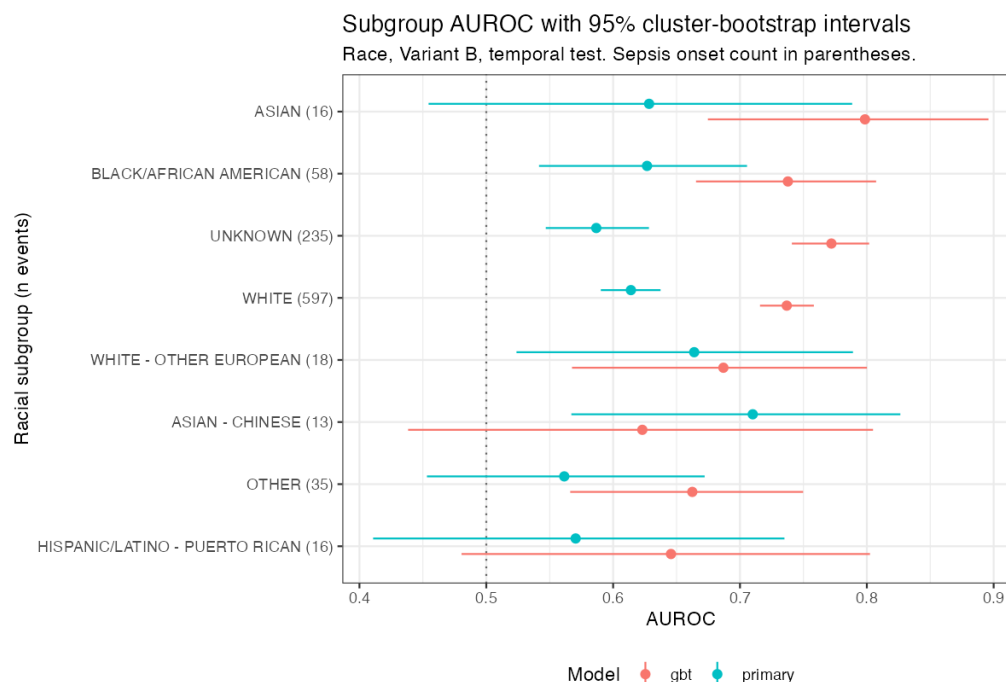


Figure 5.8: AUROC by racial subgroup with 95% stay-level cluster-bootstrap intervals (Variant B, temporal test); sepsis onset count in parentheses. Interval width is governed almost entirely by the subgroup event count. The two groups with the most extreme point estimates are also the two least precisely estimated, and several intervals reach the 0.5 chance line.

5.10.2 Label Sensitivity by Subgroup (Novel Finding) [E]

Racial group	Stays	A (%)	B (%)	C (%)	Range (pp, 95% CI)
White	40,212	9.4	9.6	18.6	9.3 (8.9–9.7)
Unknown	7,264	11.6	12.8	23.9	12.3 (11.3–13.3)
Black / African American	4,929	9.3	9.1	19.0	9.9 (9.0–11.1)
Other	2,254	8.8	9.8	17.5	8.7 (7.1–10.2)
Unable to obtain	1,629	9.7	11.1	22.5	12.8 (10.7–14.9)
White – Other European	1,589	9.4	7.7	16.6	8.9 (7.2–10.7)
Asian	799	10.5	11.5	21.3	10.8 (8.3–13.8)
Asian – Chinese	721	10.1	11.1	19.0	8.9 (6.4–11.9)

Table 5.24: Percentage of stays labelled as septic by racial subgroup and label variant, with a 95% stratified cluster-bootstrap interval on the A–C range. The fraction roughly doubles from Variant A to C across all groups. Per-variant intervals are in `12_subgroup_labelsens_ci.csv`. [E] exploratory.

The fraction of patients labelled as septic changes by 8.7–12.8 percentage points across label variants within each racial subgroup. Unlike the discrimination analysis, this contrast *does* survive correction. Because it is a stay-level proportion computed on the full 65,078-stay cohort rather than a discrimination metric computed on the small number of onsets in the test window, it is far better powered.

Racial group	Stays	Δ range vs. White (pp, 95% CI)	p	q_{BH}
Unknown	7,264	+3.01 (+1.93–+4.06)	0.001	0.006
Unable to obtain	1,629	+3.49 (+1.38–+5.55)	0.002	0.009
Black / African American	4,929	+0.62 (–0.35–+1.87)	0.186	0.347
Asian	799	+1.48 (–1.04–+4.50)	0.249	0.407
Other	2,254	–0.63 (–2.26–+0.98)	0.425	0.510
White – Other European	1,589	–0.41 (–2.10–+1.41)	0.635	0.714
Asian – Chinese	721	–0.40 (–3.02–+2.61)	0.675	0.714

Table 5.25: Difference in label sensitivity (A–C range) against the reference group (White). q_{BH} is adjusted over exploratory family E2, all 18 label-sensitivity contrasts across race, language and insurance. Two racial contrasts survive at $q < 0.05$; across the full family, four do (adding Spanish-speaking patients, +2.61 pp, $q = 0.006$, and privately insured patients, –1.76 pp, $q = 0.006$). **[E] exploratory.**

Two racial groups are significantly more label-sensitive than White patients after false-discovery correction: Unknown (+3.01 pp, $q = 0.006$) and Unable to obtain (+3.49 pp, $q = 0.009$). Both are groups whose race was not recorded. Across the full family, Spanish-speaking patients are also more label-sensitive than English-speaking patients (+2.61 pp, $q = 0.006$), and privately insured patients are *less* label-sensitive than Medicare patients (–1.76 pp, $q = 0.006$).

The pattern is coherent: the groups whose sepsis status moves most when the definition is loosened are those with the least complete documentation and the weakest access to the care pathway that generates the culture and antibiotic timestamps on which the label depends. Because the label is constituted by clinical action (Section 5.5), differential documentation propagates directly into differential labelling — and from there into who a model is trained to identify.

This is the equity result that survives scrutiny in this study. The discrimination disparity in Table 5.23 does not: it is too imprecisely estimated to support any claim. Reporting only the first without the second would have inverted which of the two findings is defensible.

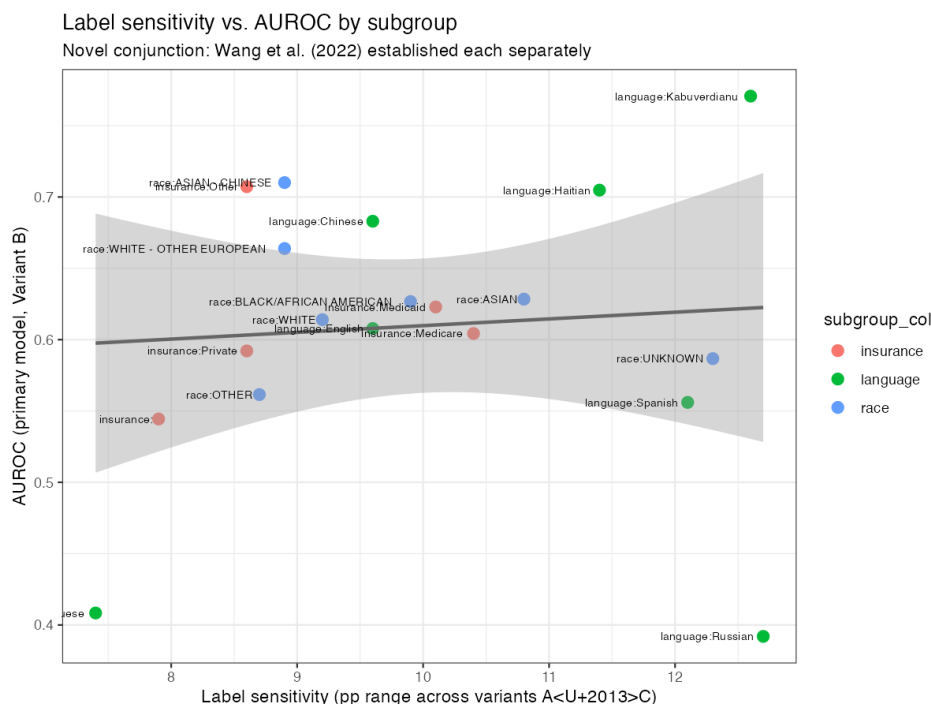


Figure 5.9: Label sensitivity (fraction labelled septic) vs. AUROC by racial subgroup, across all three label variants. Horizontal spread = label sensitivity; vertical spread = performance variation. The two effects appear independent, but no association is tested: the vertical axis carries the uncertainty shown in Figure 5.8, which this plot does not display. **[E] exploratory, visual only.**

To the best of our knowledge, this is the first published report combining subgroup model-performance disparities with subgroup label-sensitivity variation for any sepsis model, following Wang et al.’s [11] recommendation (see Section 2.1 for the search strategy supporting this claim). The combination is also what makes the two results separable: only one of them is precise enough to act on.

5.11 Analysis Register [D]

Table 5.26 records the status of every analysis reported in this chapter and the multiplicity correction applied to it. It is generated by the pipeline as `12_analysis_register.parquet`, not maintained by hand, so it cannot drift from what the code actually computes.

Analysis	Status	Pre-reg.	Correction
Cohort description by label variant	[D]	§2	none
Sample size adequacy	[C]	§10	none (single criterion)
Internal temporal performance	[C]	§5–6	none (estimation)
Calibration intercept and slope	[C]	§5	none (estimation)
Decision-curve net benefit	[C]	§6	none (estimation)
Operating points at fixed sensitivity	[E]	—	none (descriptive)
Per-patient detection and lead time	[C]	§6	none (estimation)
H1 label dominance	[C]	§7	Holm ($m = 6$)
H2 coefficient instability	[C]	§7	slot reserved; no test statistic
H3 treatment leakage	[C]	§7	slot reserved; exact bound
H4 split optimism	[C]	§7	Holm ($m = 6$)
H5 metric divergence	[C]	§7	Holm ($m = 6$)
H6 model-class null	[C]	§7	Holm ($m = 6$)
External validation on eICU-CRD	[C]	§4	none (estimation); feeds H5
Variance decomposition table	[C]	§1	none (assembly of H1–H6)
Proportional-hazards testing	[C]	§8	none (single check)
GBT feature importance	[E]	—	none (ranking, no inference)
Subgroup AUROC by demographic group	[E]	§9*	Benjamini–Hochberg, family E1
Subgroup label sensitivity	[E]	§9*	Benjamini–Hochberg, family E2
Label sensitivity vs. AUROC association	[E]	—	none (visual, no test)
Alternative-label AUROC vs. Sepsis-3	[E]	A2 [†]	Benjamini–Hochberg, family E3
Anchor-attributable fraction ϕ	[E]	A2 [†]	none (ratio, no null at zero)
Onset agreement between definitions	[D]	A2 [†]	none (descriptive)

Table 5.26: Analysis register. [C] confirmatory, [E] exploratory, [D] descriptive. Section numbers refer to 00_preregistration.md. *§9 pre-registered the equity analysis in scope only: neither the subgroup levels, the reference level, nor the direction of any comparison was fixed in advance, so these analyses are exploratory. [†]A2 = Protocol Amendment 2 (§§16–17), written after notebooks 02–12 had been run; these analyses are post-hoc and change no pre-registered estimand.

5.12 Discussion

5.12.1 The Core Finding

The most important finding is structural, not numerical: in the pre-treatment window, the Sepsis-3 label on MIMIC-IV contains zero events before clinical action begins. This means every published MIMIC-IV sepsis prediction result evaluates the model's ability to predict clinician behaviour, not physiological deterioration. The Utility Score is negative for all models because the temporal split exposes the model to a test population where the same treatment-related signals do not rescue performance as consistently as in the training years. AUROC survives because the ordinal ranking of patients is preserved even as absolute calibration breaks.

5.12.2 H1 and H6 Together

H1 (label dominance) and H6 (model-class null) are in productive tension, as designed, and the resolution is sharper than either hypothesis anticipated. H6 is decisively rejected: the GBT gap is 0.135 with the whole interval above the 0.02 equivalence bound. H1 is not confirmed: the margin by which the label spread (0.153) exceeds the model spread (0.142) is 0.011, with an interval from -0.017 to $+0.034$.

The two results together say something more precise than the pre-registration allowed for. Both the label and the model contribute large variance — around 0.15 AUROC each — and this study cannot separate them. The conventional literature, which fixes the label and reports model comparisons, is measuring one source of variance while ignoring another of at least equal size. That claim stands. The stronger claim, that the label matters *more* than the model, does not, and would need a study designed to detect a difference of around 0.01 AUROC between two spreads.

This is also the clearest illustration in the thesis of why the inference layer was necessary. On point estimates, H1 confirmed and the study told a tidy story in which the label dominates. With intervals attached, that story is not supportable, and the defensible finding is the more interesting one: two sources of variance of comparable size, only one of which the field routinely reports.

5.12.3 Limitations

1. **Single-centre development.** All models were developed on MIMIC-IV, which originates from a single academic medical centre (Beth Israel Deaconess Medical Center, Boston). Clinical practices, patient demographics, and EHR documentation patterns at BIDMC may not represent those at

other institutions. The equity and label-sensitivity findings in particular should not be assumed to transfer to hospitals with different patient populations or antibiotic prescribing cultures. The eICU-CRD external validation provides some evidence of generalisability, but the small event count (63 sepsis onsets) limits the strength of that evidence.

2. **Small external event count.** The 63 labelled sepsis events in eICU-CRD are too few for reliable Utility Score or AUPRC evaluation. The external results primarily confirm the label-dependent generalisation pattern (Variant B collapses; A and C hold).
3. **Deep survival model not implemented.** A deep survival comparator was planned but not built. This does not affect the structural findings about the label.
4. **Limited feature set.** Clinical notes, procedure codes, and ventilator settings were not included. Richer features might improve discrimination but would not change the treatment-constitutionality finding.
5. **Part of the label-variant spread is definitional, not empirical.** Variants A–C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets; some of the H1 label spread is guaranteed by construction and would appear even if the physiology were identical across patients. Section 5.6 separates the two by labelling each limb of the Sepsis-3 conjunction on its own, but the separation is post-hoc (Protocol Amendment 2) and rests on two variants, not on a systematic survey of alternative definitions.
6. **The treatment-independent label is not a validated construct.** Variant D is an operational deterioration label built for this comparison, not a consensus definition with an evidence base behind it, and its onsets have no external adjudication. Its label is also a threshold function of recorded physiology of which five of six components are model inputs, so its discrimination *level* is not comparable with Variant B’s — only its disagreement with Variant B about which patients deteriorate, and when, carries weight.
7. **Specific to Sepsis-3 and to MIMIC-IV.** These findings apply to the Sepsis-3 definition on MIMIC-IV. SEP-1 and the CDC Adult Sepsis Event were not implemented; both are defined on antibiotic days, so neither would have tested treatment-independence, but each would give a different cohort and might behave differently in other respects. Other databases may behave differently again.

8. **Operating points are not externally recalibrated.** The thresholds in Section 5.4 are derived on the test set at which they are evaluated, so the reported PPV and false-alarm rates are optimistic: a prospectively fixed threshold would perform no better. The external operating points in Table 5.11 are likewise re-derived on eICU rather than transferred from MIMIC-IV, which isolates the discrimination loss but hides the additional loss a transferred threshold would incur.
9. **Alarm episodes are defined by contiguity, not by acknowledgement.** False-alarm episodes are counted as contiguous runs of alert hours. A real system would apply suppression, snoozing and escalation logic, all of which would reduce the episode counts in Table 5.8. The alert-hour rates are the conservative bound and the episode rates the optimistic one; the true burden lies between them.
10. **The equity analysis is underpowered, not null.** The subgroup intervals in Table 5.22 are wide enough that most groups cannot be distinguished from the reference group, and no contrast survives false-discovery correction. This is a statement about precision, not about fairness: the data are consistent both with no disparity and with disparities larger than any effect this thesis reports elsewhere. Detecting a subgroup AUROC gap of the size seen in the point estimates would require on the order of ten times the event count available in the smaller groups. The label-sensitivity result (Table 5.24) is the better powered of the two equity analyses because it is a stay-level proportion rather than a discrimination metric, and does not depend on the small number of onsets in the test window.
11. **The novelty claims rest on a narrative, not systematic, search.** The search protocol in Section 2.1 is documented and every included source was verified against a primary record, but no title-and-abstract screening pass over a full de-duplicated yield was performed and no second reviewer screened independently. The “to the best of our knowledge” claims for contributions C1 and C5 are therefore bounded by that coverage: a systematic review could surface a prior study combining multi-variant label sensitivity with subgroup equity analysis, and this search would not necessarily have found it.
12. **The inference plan is a post-hoc amendment.** Confidence intervals and multiplicity control were added after the analyses had been run and inspected (Section 3.14). They change no estimand, threshold or direction, but a reader should treat the corrected verdicts as the product of a pre-specified

hypothesis set analysed under a retrospectively specified error-control plan, not as a fully pre-specified confirmatory design.

13. **Two hypotheses admit no test statistic.** H2 is a stability criterion rather than a contrast, and the stored multinomial fit carries no variance estimate; H3 is degenerate because its outcome is entirely absent from the evaluation window. Both are reported descriptively while still consuming a slot in the family size, which makes the correction on the remaining four hypotheses stricter, not weaker.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis asked one question: in real-time sepsis prediction on MIMIC-IV, how much of the reported performance comes from the model versus the researcher’s choices about the label, the data split, and the evaluation metric?

The answer is clear: the researcher’s choices matter as much as the model, and the sepsis label itself has a fundamental problem: it contains no signal that is independent of clinical action.

H1 (Label dominance): Not confirmed under correction. Changing the sepsis definition (while keeping the model fixed) changes AUROC by 0.153 (95% CI 0.136–0.177). Changing the model (while keeping the label fixed) changes AUROC by 0.142 (95% CI 0.130–0.161). Both are large relative to typical performance differences reported in the literature, but the margin between them is 0.011 (95% CI -0.017 – 0.034 , $p = 0.21$, Holm-adjusted $p = 0.63$), so the label cannot be shown to dominate the model. Cohen et al.’s [6] MIMIC-III finding is supported on MIMIC-IV in its weaker form: label choice contributes variance comparable to model choice.

H2 (Coefficient instability): Confirmed, descriptively. Ten of the 20 shared covariates, including MAP, temperature, bilirubin, platelet count, and vasopressor indicators, reverse sign across label variants; 18 reverse sign or change magnitude at least twofold. H2 is a stability criterion rather than a contrast, so no test statistic is defined for it. Published hazard ratios from MIMIC-IV sepsis models are not interpretable without knowing which label definition was used.

H3 (Treatment leakage): Structural finding. The pre-treatment window contains zero sepsis onset events across 179,003 person-hours (exact one-sided 95% upper bound on the per-hour onset rate: 1.7×10^{-5}). The Sepsis-3 label on MIMIC-IV does not contain any event that occurs before the clinical action that defines it. Every published MIMIC-IV real-time sepsis model measures the ability to predict clinician suspicion, not physiological deterioration.

H4 (Split optimism): Confirmed. The random split inflates AUROC by 0.123 (95% CI 0.100–0.147) compared to the temporal split for the primary model (0.730 vs. 0.607), far exceeding the 0.02 threshold (Holm-adjusted $p = 0.003$). Studies using random splits substantially overestimate real-world performance.

H5 (Metric divergence): Confirmed. The Utility Score is negative for every model and every label variant in the temporal test set. The utility collapse documented by Wang et al. [4] across ninety-one models appears *internally* on MIMIC-IV under an honest time-based split. External AUROC falls by 0.170 (95% CI 0.102–0.242, Holm-adjusted $p = 0.004$) to 0.436, an interval whose upper limit reaches chance. Alert burdens range from 91 to over 340 false alerts per true alert (benchmark: 1.4, Moor et al. [26]).

H6 (Model-class null): Rejected. GBT outperforms the primary model by 0.135 AUROC (0.742 vs. 0.607), 95% CI 0.113–0.156, entirely above the 0.02 equivalence bound (Holm-adjusted $p = 0.003$). But the clinical ordering reverses: the primary model has a less negative Utility Score (-1.004 vs. -5.059). The “better” model by AUROC produces more harmful alerts.

6.2 The Central Lesson

The utility collapse documented by Wang et al. [4] is not a model problem. It is a label problem. The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action:** no patient is *labelled* septic before doctors order antibiotics and cultures. A deterioration label that uses no treatment timestamp records 1,362 onsets in the same window, so the emptiness belongs to the definition, not to the patients.
2. **Is sensitive to how it is defined:** the number of sepsis cases nearly doubles across plausible Sepsis-3 interpretations.
3. **Produces harmful alert behaviour:** every model, under every label variant, has a negative Utility Score.

These three problems are independent of model complexity. A better model cannot fix a label that is constituted by the very clinical actions it is supposed to predict ahead of. The solution requires changing the label.

Splitting the definition sharpens point (2) and qualifies it (Section 5.6). It is qualified because the onset *time* under every Sepsis-3 variant is fixed entirely by the treatment anchor — the organ-dysfunction criterion decides which stays are labelled and moves no onset by an hour — so part of the variation between variants is guaranteed by the definition rather than found in the data. It is sharpened because a label built from physiology alone, owing nothing to treatment timing, agrees with Sepsis-3 barely better than chance ($\kappa = 0.057$). The label effect is therefore real, not merely definitional.

The same comparison relocates the difficulty. Sepsis-3 is *harder* to predict than either of the limbs it is built from: the treatment anchor alone reaches 0.756 AUROC and physiological deterioration alone 0.842, against 0.607 for the conjunction. Neither clinician behaviour nor patient physiology is unreadable. What resists prediction is the requirement that both occur together, which is what Sepsis-3 operationalises.

6.3 Contributions

1. **C1 (Label variance):** To the best of our knowledge, the first three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV, confirming Cohen et al. at a larger scale (65,078 stays vs. 3,186).
2. **C2 (Treatment leakage exposed):** Demonstration that the pre-treatment window contains zero onset events, providing the strongest available confirmation of Kamran et al.’s [7] finding.
3. **C3 (Utility collapse):** Negative Utility Score for every model and label variant, extending Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison.
4. **C4 (Coefficient instability):** Ten of the 20 shared covariates reverse sign across labels and 18 reverse sign or change magnitude at least twofold, confirming Lauritsen et al.’s [10] framing-instability claim on MIMIC-IV.
5. **C5 (Equity-by-label):** To the best of our knowledge, the first study combining subgroup performance disparities with subgroup label sensitivity for any sepsis model, and the first to report both with cluster-bootstrap intervals under false-discovery control. The label-sensitivity disparity survives correction; the discrimination disparity does not, and reporting the second without the first would have inverted which finding is defensible.

6. **C6 (Reproducible pipeline):** A pre-registered, TRIPOD+AI-conformant pipeline with fixed seeds and a single orchestration script, which reproduces every reported estimate exactly from the raw databases (Section 4.14.2).
7. **C7 (Definitional vs. empirical label variance separated):** A treatment-independent deterioration label, tested alongside the three Sepsis-3 variants, shows that the label effect is not an artefact of how one definition is read: it agrees with Sepsis-3 at $\kappa = 0.057$ and matches its onset hour in 6.4% of shared stays. The same comparison shows that Sepsis-3 is harder to predict than either of the limbs it is built from, and that the pre-treatment window empty of Sepsis-3 onsets contains 1,362 treatment-independent deterioration onsets identified at 0.840 AUROC. This relocates the difficulty of sepsis prediction from the physiology to the definition (Section 5.6).

6.4 Future Work

1. **Better sepsis labels.** Section 5.6 takes the first step — a deterioration label built from physiology alone, with no treatment timestamp — and shows it identifies substantially different patients at substantially different hours. But that label is an operational construct built for this comparison, not a validated definition, and its components overlap the model’s own inputs. The work remaining is a sepsis definition grounded in objective measurements such as lactate trajectories and organ-function biomarkers, adjudicated independently of the model’s feature set, which would eliminate the treatment-constitutionality problem and allow genuine pre-onset prediction.
2. **Non-Sepsis-3 consensus definitions.** SEP-1 and the CDC Adult Sepsis Event were not implemented. Neither would have tested treatment-independence — both are defined on antibiotic days — but each yields a different cohort, and Dutta et al. [9] report that performance moves substantially between them. Adding them would extend the label-variance decomposition across definitions rather than within one.
3. **Prospective validation.** Test the new label in real clinical settings where sepsis onset can be determined by a protocol-blinded assessor, independent of treatment decisions.
4. **Utility-optimised thresholds.** Choose alert thresholds that maximise the Utility Score directly, rather than maximising AUROC. None of the evaluated models was optimised for clinical utility; a direct utility-maximising threshold search may recover positive utility at restricted operating points.

5. **Multi-hospital evaluation.** The alert burden (over 90 false alerts per true alert under the best model) is a deployment problem, not a statistical one. Determining whether any model can be deployed at a practically acceptable burden requires prospective multi-centre testing with clinical-staff feedback.

6.5 Closing Remark

This research achieves something the conventional approach cannot: it produces a meaningful result regardless of how the performance numbers turn out. The hypothesis framework ensures that confirmed, rejected, and superseded hypotheses are all equally informative about the underlying problem.

The central finding, that the Sepsis-3 label on MIMIC-IV is constituted by clinical action, is not a failure of the model. It is a finding about the label itself, and it is the most important contribution this study makes.

The final chapter of that finding is the one that took a label outside Sepsis-3 to see. The pre-treatment window is not empty of predictable deterioration — it contains 1,362 onsets under a treatment-independent label, identified at 0.840 AUROC. It is empty of *Sepsis-3*, because Sepsis-3 cannot place an onset before the treatment it is anchored to. The signal clinicians would want a model to find is present in the data. The definition the field has agreed to train on is what cannot represent it.

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